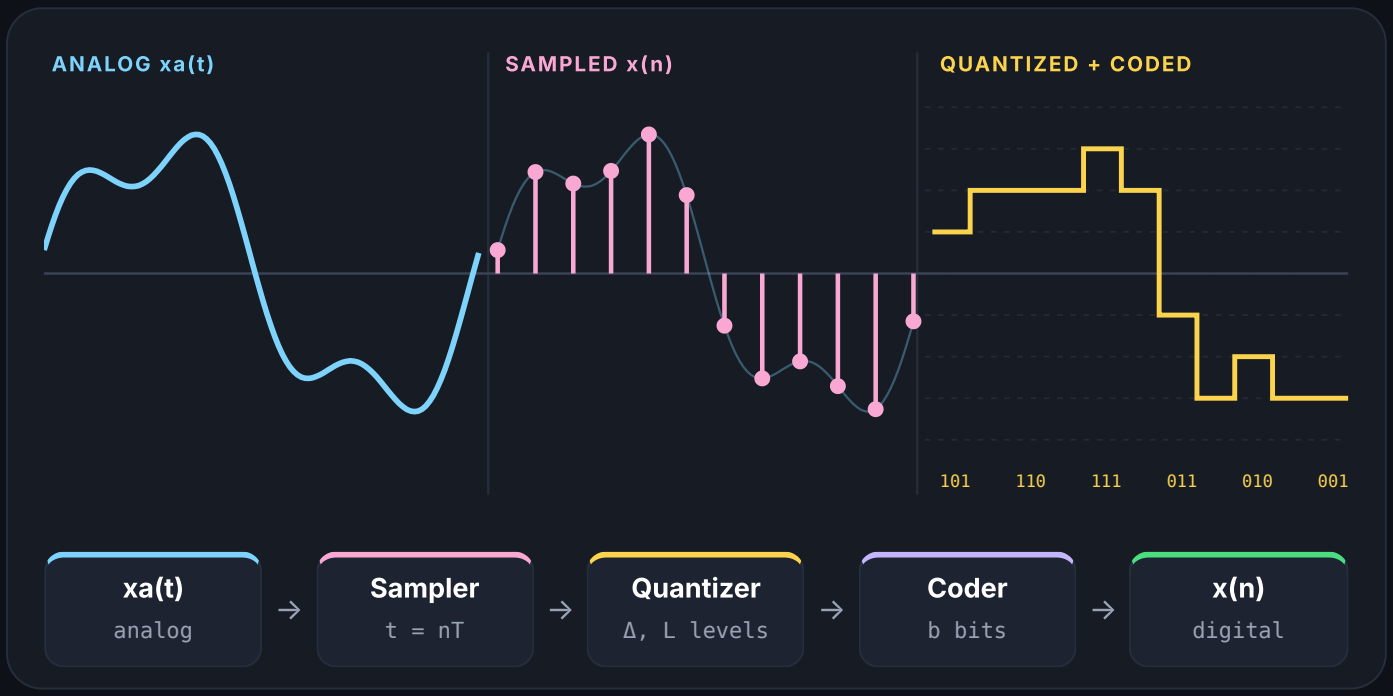
01

CHAPTER

Introduction to Signal Processing

The Complete Chapter 1 Pack

Theory and problems from the lectures, previous exams and Sheet 1 — solved the doctor's way



22	18	8	25
theory questions	solved problems	Sheet 1 questions	previous exams & quizzes

PART A
Theory
Lectures 2 · 3 · 4

PART B
Problems
Method · Examples · Exams

PART C
Sheet 1
Theory · Problems

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DSP / ch01 / contents

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Theory · Problems · Sheet 1 · 49 questions

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How to study from this file

A: theory, most repeated first. B: start with the solving method, then examples and exams. C: all of Sheet 1.

DSP / ch01 / repeated

Most Repeated Theory Questions

Most repeated in previous exams — start with these

Times asked	Question
9	Q9. In detail, what are the main characteristics of the discrete-time sinusoidal signals?
8	Q18. In A/D, explain in detail the quantization process of continuous-amplitude signals.
8	Q15. Find the relationship between the frequency of the analog signal f, the frequency of the discrete-time signal F, and the sampling frequency Fs.
6	Q16. In sampling process, derive the sampling condition $Fs \geq 2f_{max}$ (avoiding aliasing).
4	Q22. Check (✓) for the right statements and (×) for the wrong statements and correct the mistakes.
4	Q14. In A/D, explain in detail the sampling process of analog signals.
4	Q13. Explain (with a block diagram) the basic three operations performed in the A/D converter.
4	Q8. In detail, what are the main characteristics of the continuous-time sinusoidal signals?
4	Q4. Explain, with drawings, the different classifications of signals.

SECTION 01

Lecture 2

Signals, Systems & Classification

Questions — most frequent first

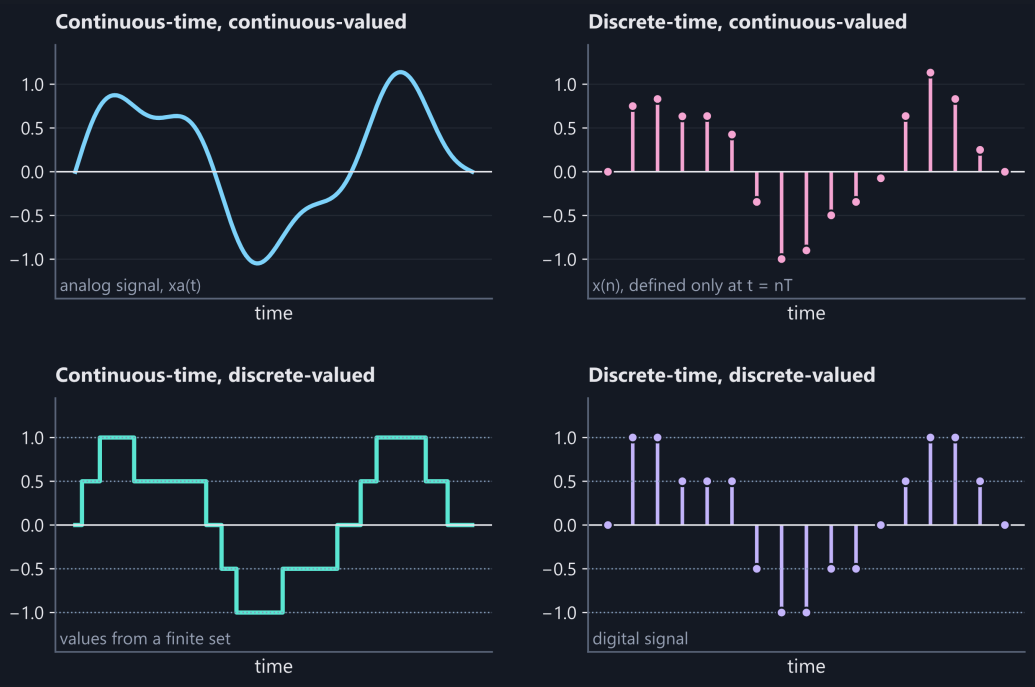
- ☐ Q4 Explain, with drawings, the different classifications of signals. p. 4
- ☐ Q1 Define the following items: Signal, System, and Signal Processing (give an example f... p. 6
- ☐ Q2 Draw the block diagram of (a) an analog signal processor and (b) a digital signal pr... p. 7
- ☐ Q3 What are the advantages of digital signal processing over analog signal processing? ... p. 8
- ☐ Q5 Compare between deterministic and random signals. p. 9
- ☐ Q6 What is a digital signal? Why must a signal be digital to be processed digitally? p. 9
- ☐ Q7 What is quantization? Compare between truncation and rounding with an example. p. 10

Q4 ● Frequent 4×

Explain, with drawings, the different classifications of signals.

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.
4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.

Lecture 2



Key idea احفظ التصنيفات كأزواج: continuous / discrete time، و continuous / discrete valued، و deterministic / random، و multichannel / multidimensional.

Key words: Deterministic vs. random

Source — from your PDFs

Lecture **LEC 2** · Sec. 1.4 · p. 18

Came in Midterm 8-12-2020 · Q2-a | Revision Sheet 1 | Sheet 1 · 1.3 | Midterm theory notes

Lecture 2

Q1 3×

Define the following items: Signal, System, and Signal Processing (give an example for each).

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: $s1(t) = 5t$ (varies linearly with t), $s2(t) = 20t^2$ (varies quadratically with t), and $s(x, y) = 3x + 2xy + 10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

Key idea الـ Signal أي كمية فيزيائية بتتغير مع متغير مستقل زي الوقت. الـ System هو الجهاز اللي بيشتغل على الـ signal. والـ Signal Processing إننا نعدّي الـ signal على system، زي الـ filter.

Key words: Signal | System | Signal processing

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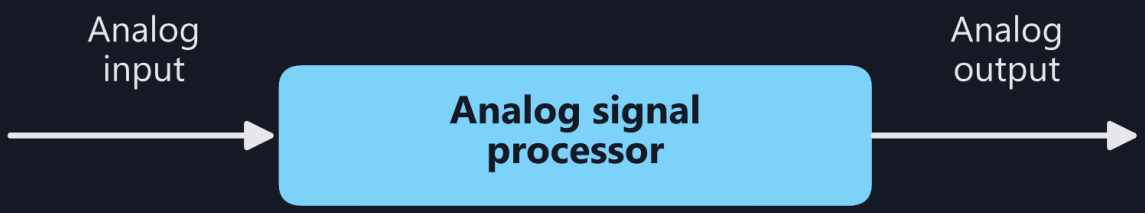
Lecture	**LEC 2** · Sec. 1.1 · p. 4
Came in	Sheet 1 · 1.1 · Midterm OLD · Model-C · Quiz 1 · Section

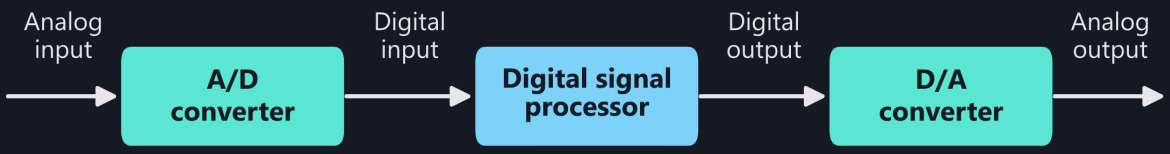
Q2 2×

Draw the block diagram of (a) an analog signal processor and (b) a digital signal processor.

There are two types of signal processing systems (Sec. 1.2):

1. **Analog signal processor (Fig. 1.1):** the analog input signal is processed directly by an analog signal processor to give an analog output signal.
2. **Digital signal processor (Fig. 1.2):** provides an alternative method for processing the analog signal. The **A/D converter** converts the analog input into a digital signal (a sequence of numbers), the **digital signal processor** performs the required operations on it, and the **D/A converter** converts the result back into an analog output signal.





Key idea في الـ analog الإشارة بتدخل الـ processor على طول. في الـ digital لازم تعدّي على الأول A/D، وبعدين الـ digital processor، وبعدين D/A عشان ترجع analog.

Source — from your PDFs

Lecture	**LEC 2** · Sec. 1.2 · p. 11
Came in	Revision Sheet 1 · Sheet 1 · 1.2

Q3 1×

What are the advantages of digital signal processing over analog signal processing? What is its main limitation?

1. **Flexibility:** a digital programmable system allows reconfiguring the operations simply by changing the program; an analog system needs hardware redesign.
2. **Accuracy:** a digital system provides much better control of accuracy requirements; tolerances in analog components make accuracy hard to control.
3. **Storage:** digital signals are easily stored on magnetic media without deterioration, so they are transportable and can be processed off-line.
4. Allows the implementation of more **sophisticated** signal processing algorithms.
5. **Precise mathematical operations** are very difficult in analog form but routine on a digital computer using software.
6. In some cases a digital implementation is **cheaper** than its analog counterpart.

Limitation: the speed of operation of A/D converters and digital signal processors; signals with extremely wide bandwidths may be beyond the state of the art of digital hardware.

Key idea الـ DSP أحسن في الـ flexibility والـ accuracy والتخزين والـ algorithms المعقدة والتكلفة. عيبه سرعة الـ A/D والـ processor، ودي اللي بتحدد الـ bandwidth.

Key words Flexibility · Accuracy · Storage

Source — from your PDFs

Lecture	**LEC 2** · Sec. 1.3 · p. 13
Came in	Midterm OLD

Q5 1×

Compare between deterministic and random signals.

- **Deterministic:** any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. All past, present, and future values are known precisely, without any uncertainty.
- **Random:** otherwise, we refer to these signals as random.

Key idea الـ deterministic نقدر نوصفها بالظبط بمعادلة أو جدول ونعرف قيمتها في أي وقت. الـ random ما نقدرش نتوقع قيمتها بالظبط.

Key words Deterministic Random

Source — from your PDFs

Lecture	**LEC 2** · Sec. 1.4.3 · p. 26
Came in	Midterm OLD · Model-A 2013/2014

Q6 Less frequent

What is a digital signal? Why must a signal be digital to be processed digitally?

- A **digital signal** is discrete in time and its values are discrete (Fig. 1.4: a digital signal with four different amplitude values).
- In order for a signal to be processed digitally, it must be discrete in time and its values must be discrete (i.e., it must be a digital signal).

Key idea الـ digital signal متقطعة في الوقت وفي الـ amplitude الاتنين، والـ processor الرقمي ما يتعاملش غير مع قيم sampled و quantized.

Lecture 2

Source — from your PDFs

Lecture **LEC 2** · Sec. 1.4.2 · p. 23

Q7 Less frequent

What is quantization? Compare between truncation and rounding with an example.

- **Quantization:** the process of converting a continuous-valued signal into a discrete-valued signal; it is basically an approximation process, accomplished by rounding or truncation.
- **Example:** if the allowable values are the integers 0 through 15, the value 8.58 is approximated by **8** by truncation, or by **9** by rounding to the nearest integer.

Key idea الـ quantization بتقرّب كل قيمة لواحد من عدد محدود من الـ levels. الـ truncation بيشيل الأرقام الزيادة، والـ rounding بياخد أقرب level.

Key words Quantization

Source — from your PDFs

Lecture **LEC 2** · Sec. 1.4.2 (truncation / rounding) · p. 25

Lecture **LEC 4** · Sec. 1.6.3 · p. 23

★ Cheat sheet · Lecture 2

Remember

Q1 Signal = physical quantity varying with an independent variable; system = device that operates on a signal; processing = passing a signal through a system (e.g. filtering).

Q2 Analog: input → analog signal processor → output. Digital: input → A/D → digital signal processor → D/A → output.

Q3 DSP wins in flexibility, accuracy, storage, sophisticated algorithms and cost; its limitation is the speed of the A/D converters and processors (bandwidth).

Q4 Continuous vs discrete time, continuous vs discrete valued, deterministic vs random, multichannel vs multidimensional.

Q5 Deterministic = described exactly by an expression, table or rule (past, present and future known); random = cannot be predicted exactly.

Q6 A digital signal is discrete in time and in amplitude; a digital processor can only handle sampled and quantized values.

Q7 Quantization maps a continuous value to a finite set of levels; truncation drops the extra digits, rounding takes the nearest level.

SECTION 02

Lecture 3

Concept of Frequency

Questions — most frequent first

- ☐ **Q9** In detail, what are the main characteristics of the discrete-time sinusoidal signals... p. 11
- ☐ **Q8** In detail, what are the main characteristics of the continuous-time sinusoidal signa... p. 13
- ☐ **Q10** Show that a discrete-time sinusoid is periodic only if its frequency F is a rational... p. 14
- ☐ **Q11** Prove that discrete-time sinusoids whose frequencies are separated by an integer mul... p. 15
- ☐ **Q12** What are harmonically related complex exponentials? Compare between the continuous-t... p. 16

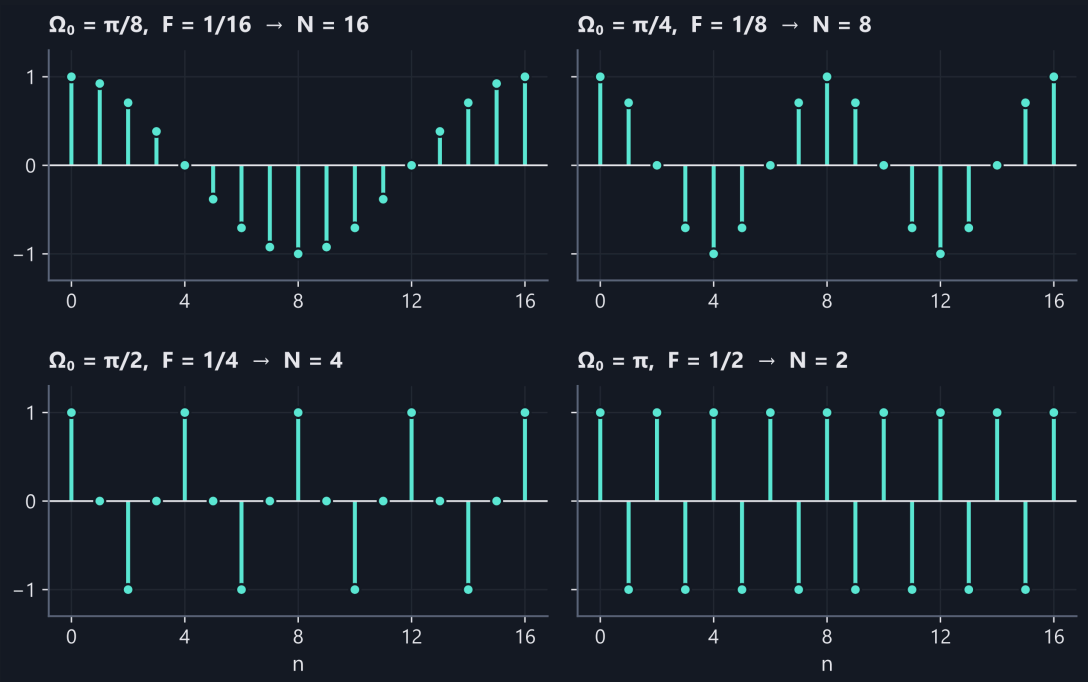
Q9 ● Frequent 9×

In detail, what are the main characteristics of the discrete-time sinusoidal signals?

$x(n) = A\cos(\Omega n + \theta) = A\cos(2\pi F n + \theta)$, where n is the sample number, Ω is in radians/sample and F in cycles/sample.

Lecture 3

1. **(a)** A discrete-time sinusoid is **periodic only if its frequency F is a rational number** (F_o = k/N).
2. **(b)** Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are **identical**. Frequencies in $-\pi \le \Omega \le \pi$ ($-1/2 \le F \le 1/2$) are unique; all frequencies $|\Omega| > \pi$ are **aliases**.
3. **(c)** The **highest rate of oscillation** is attained when $\Omega = \pi$ (or $\Omega = -\pi$), i.e. F = 1/2 (or F = −1/2).



Key idea الـ discrete sinusoid تبقى periodic بس لو F rational. الترددات اللي بينها 2π بتدّي نفس الإشارة، وأسرع تذبذب بيكون عند ω = π.

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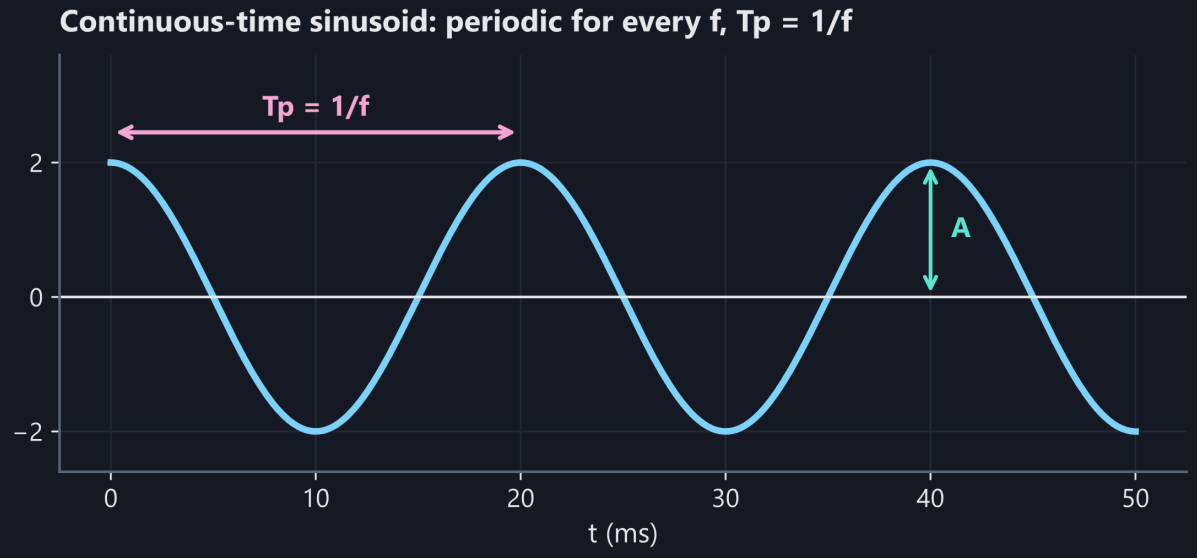
Lecture	**LEC 3** · Sec. 1.5.2 · p. 9
Came in	Final 11-6-2023 · Final 7-6-2023 · Q4-a · Final 12-1-2019 · Final 2-1-2017 · Final 23-12-2017 · Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm OLD · Final theory notes

Q8 ● Frequent 4×

In detail, what are the main characteristics of the continuous-time sinusoidal signals?

$x_a(t) = A\cos(\omega t + \theta) = A\cos(2\pi f t + \theta)$, where A is the amplitude, $\omega = 2\pi f$ the frequency in rad/s, f in Hz, and θ the phase in radians.

1. **(a)** For every fixed value of the frequency f, $x_a(t)$ is **periodic**: $x_a(t + T_p) = x_a(t)$, where $T_p = 1/f$ is the fundamental period.
2. **(b)** Continuous-time sinusoidal signals with distinct (different) frequencies are themselves **distinct**.
3. **(c)** Increasing the frequency f results in an **increase in the rate of oscillation** of the signal; f can be increased without limit.



Key idea الـ continuous sinusoid دايمًا periodic، وكل تردد مختلف بيدّي إشارة مختلفة، وكل ما التردد يزيد الإشارة بتتذبذب أسرع.

Key words periodic

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.1 · p. 3

Came in Final 20-1-2025 · Q3-a | Final 9-6-2026 · Q1-a | Quiz 1 Summer 26 · Q1-b | Midterm OLD

Q10 2×

Show that a discrete-time sinusoid is periodic only if its frequency F is a rational number. How do you find its fundamental period?

By definition, x(n) is periodic with period N (N > 0) if and only if x(n + N) = x(n) for all n (1.15). For a sinusoid with frequency F_o to be periodic:

$$\cos[2\pi F_0(N + n) + \theta] = \cos(2\pi F_0 n + \theta)$$

This relation is true if and only if there exists an integer k such that $2\pi F_o N = 2k\pi$, or equivalently:

$$F_0 = k / N \quad (1.16)$$

So F_o must be rational. To find the fundamental period, cancel common factors so that k and N are relatively prime; then the fundamental period is N.

Key idea عشان x(n + N) = x(n) لازم 2πFN = 2πk، يعني F = k/N لازم تبقى rational، وN بعد ما نختصر الكسر هي الـ fundamental period.

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.2 · p. 9

Came in Midterm 1 Model-2 · Q2-b | Notebook / Section

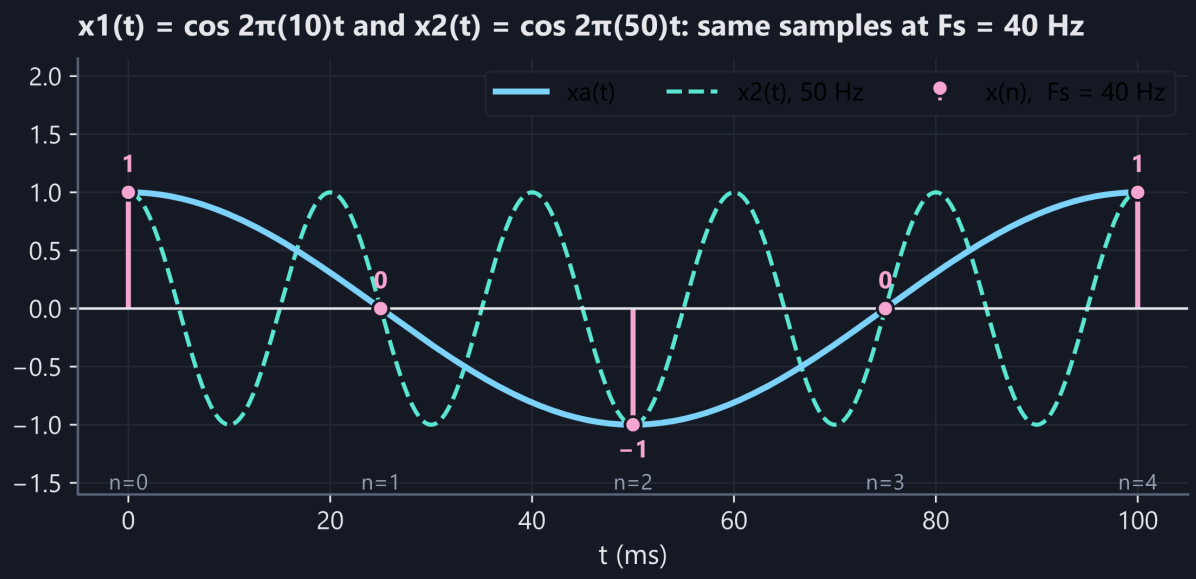
Q11 Less frequent

Prove that discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical. What is an alias?

$$\cos[(\Omega_0 + 2\pi)n + \theta] = \cos(\Omega_0 n + 2\pi n + \theta) = \cos(\Omega_0 n + \theta) \quad (1.17)$$

As a result, all sinusoidal sequences with $\Omega k = \Omega_o + 2k\pi$, $-\pi \le \Omega_o \le \pi$, are indistinguishable (identical).

Alias: any sequence from a sinusoid with $|\Omega| > \pi$ ($|F| > 1/2$) is identical to a sequence obtained from a sinusoid with $|\Omega| < \pi$; it is called an alias of the corresponding sinusoid.



Key idea لأن n عدد صحيح، $\cos((\omega 0 + 2\pi k)n) = \cos(\omega 0\ n)$. يعني الترددات اللي بينها 2πk بتدّي نفس الـ samples، وكل واحد فيهم alias للتاني.

Key words Alias

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.2 (aliases) · p. 17

Lecture 3

Q12 Less frequent

What are harmonically related complex exponentials? Compare between the continuous-time and discrete-time cases.

Sets of periodic complex exponentials with fundamental frequencies that are multiples of a single positive frequency.

	Continuous time	Discrete time
Basic signals	$s_k(t) = e^{jk\Omega_o t}$, k = 0, ±1, ±2, ...	$s_k(n) = e^{j2\pi kn/N}$, k = 0, 1, ..., N − 1
Distinct members	all distinct (infinite)	only N distinct: $s_{k+N}(n) = s_k(n)$
Common period	$T_p = 1/F_o$	N samples
Fourier series	$x_a(t) = \Sigma\, c_k\, e^{jk\Omega_o t}$	$x(n) = \Sigma_{k=0..N-1}\, c_k\, e^{j2\pi kn/N}$

Key idea في الـ continuous time فيه عدد لا نهائي من الـ harmonics المختلفة. في الـ discrete time فيه N بس مختلفين، لأن s(k+N) = s(k).

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.3 · p. 21

★ Cheat sheet · Lecture 3

Remember

Q8 $x_a(t) = A\cos(\Omega t + \theta)$ is always periodic ($T_p = 1/F$), distinct F give distinct signals, and a higher F means faster oscillation.

Q9 $x(n) = A\cos(\omega n + \theta)$ is periodic only for rational f, frequencies 2π apart are identical, and the fastest oscillation is at $\omega = \pi$ ($f = 1/2$).

Q10 $x(n + N) = x(n)$ needs $2\pi fN = 2\pi k$, so $f = k/N$ must be rational; N (with k/N in lowest terms) is the fundamental period.

Q11 $\cos((\omega_0 + 2\pi k)n) = \cos(\omega_0 n)$ for integer n, so sinusoids $2\pi k$ apart are the same; any $\omega_k = \omega_0 + 2\pi k$ is an alias of ω_0.

Q12 s_k = e^{j2πkF0 t}: infinitely many distinct ones in continuous time, but only N distinct ones in discrete time (s_{k+N} = s_k).

Equations

$$F_0 = k / N \quad (1.16)$$

$$\cos[(\Omega_0 + 2\pi)n + \theta] = \cos(\Omega_0 n + 2\pi n + \theta) = \cos(\Omega_0 n + \theta) \quad (1.17)$$

SECTION 03

Lecture 4

A/D and D/A Conversion

Questions — most frequent first

- ☐ **Q15** Find the relationship between the frequency of the analog signal f, the frequency of... p. 18
- ☐ **Q18** In A/D, explain in detail the quantization process of continuous-amplitude signals. p. 19
- ☐ **Q16** In sampling process, derive the sampling condition Fs ≥ 2f max (avoiding aliasing). p. 20
- ☐ **Q13** Explain (with a block diagram) the basic three operations performed in the A/D conve... p. 21
- ☐ **Q14** In A/D, explain in detail the sampling process of analog signals. p. 22
- ☐ **Q22** Check (✓) for the right statements and (×) for the wrong statements and correct the ... p. 23
- ☐ **Q17** State and explain the sampling theorem. Define the Nyquist rate. p. 24
- ☐ **Q19** In A/D conversion, explain in detail the coding process. p. 24
- ☐ **Q20** Why is the sampling process invertible while the quantization process is not? p. 25
- ☐ **Q21** Explain the D/A conversion process. Compare between ideal interpolation, zero-order ... p. 26

Lecture 4

Q15 · Frequent 8×

Find the relationship between the frequency of the analog signal f, the frequency of the discrete-time signal F, and the sampling frequency Fs.

Consider xa(t) = A cos(2πft + θ). Sampling at Fs = 1/T (t = nT = n/Fs):

$$x(n) = x_a(nT) = A\cos(2\pi fnT + \theta) = A\cos(2\pi nf/F_s + \theta) \quad (1.32)$$

Comparing with x(n) = A cos(2πFn + θ) (1.14):

$$F = f / F_s \quad (1.33) \qquad \text{or} \qquad \Omega = \omega T \quad (1.34)$$

Key idea لما نعوّض t = nT بنلاقي F = f/Fs، يعني التردد الرقمي هو التردد الأنالوج مقسوم على الـ sampling rate.

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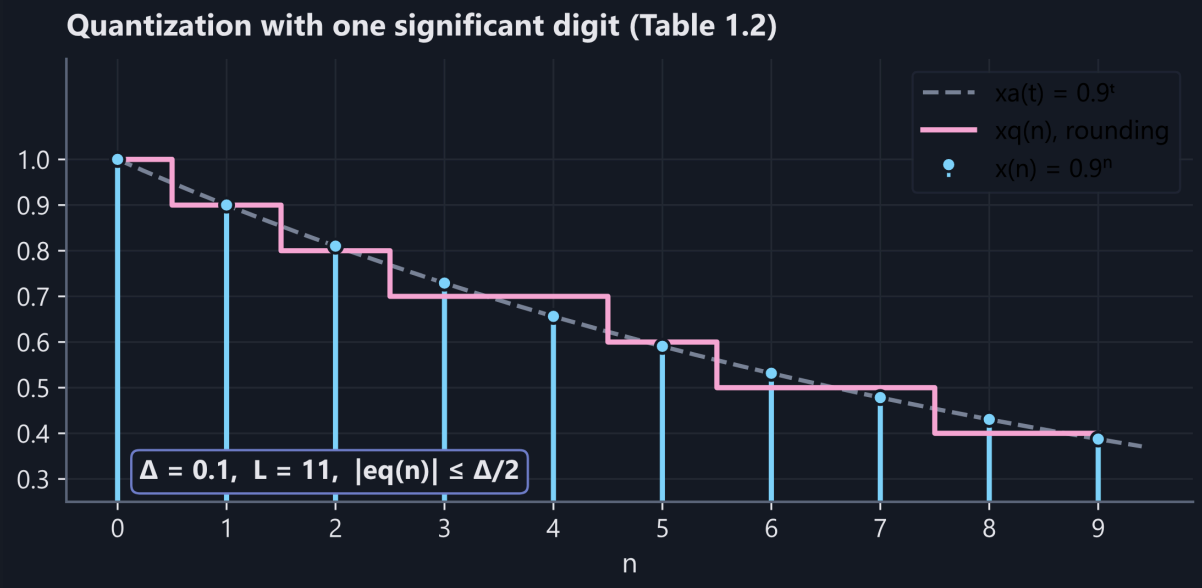
Lecture **LEC 4** · Sec. 1.6.1 · p. 8

Came in Final 18-1-2024 · Q1-a · Final 12-1-2019 · Q2-a · Final 2-1-2017 · Q2-a · Final 23-12-2017 · Q2-a · Final 28-12-2019 · Q1-a · Midterm 12-11-2025 · Q1-a · Midterm 2 · Q2-a · Final 8-9-2021

Q18 ● Frequent 8×

In A/D, explain in detail the quantization process of continuous-amplitude signals.

- **Quantization:** the process of converting a discrete-time continuous-amplitude signal into a digital signal by expressing each sample value as a finite number of digits.
- **Quantization error:** $e_q(n) = x_q(n) - x(n)$ (1.53), where $x_q(n) = Q[x(n)]$.
- The values allowed are the **quantization levels**; the distance between two successive levels is the **step size Δ**.
- In rounding: $-\Delta/2 \le e_q(n) \le \Delta/2$ (1.54), and $\Delta = (x_{max} - x_{min})/(L - 1)$ (1.55).
- Increasing the number of levels L decreases Δ, so the quantization error decreases and the accuracy increases.



Key idea كل sample بيتقرّب لواحد من L levels، والمسافة بينهم Δ = (xmax − xmin)/(L − 1)، وأقصى error مع الـ rounding هو Δ/2.

Key words Quantization Quantization error

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6.3 · p. 23

Came in Final 7-6-2023 · Q1-a | Final 17-8-2026 · Q1-a | Final 4-3-2021 · Q1-a | Final 8-9-2021 · Q1-a | Midterm 2 · Q1-b | Quiz 1 · Section | Midterm OLD | Final theory notes

Q16 ● Frequent 6×

In sampling process, derive the sampling condition $Fs \geq 2f_{max}$ (avoiding aliasing).

Since the discrete-time frequency must satisfy $-1/2 \leq F \leq 1/2$ and $F = f/Fs$:

$$-1/2 \leq f/Fs \leq 1/2 \rightarrow -Fs/2 \leq f \leq Fs/2 \quad (1.37)$$

$$f_{max} = Fs/2 \rightarrow \mathbf{Fs \geq 2f_{max}}$$

Thus to avoid the problem of aliasing, Fs is selected so that $Fs > 2f_{max}$ (1.47).

Key idea عشان كل component تتمثل صح لازم $|F| \leq 1/2$، يعني $f \leq Fs/2$، يعني لو $Fs \geq 2fmax$. ده ما حصلش، الترددات العالية بتعمل aliasing.

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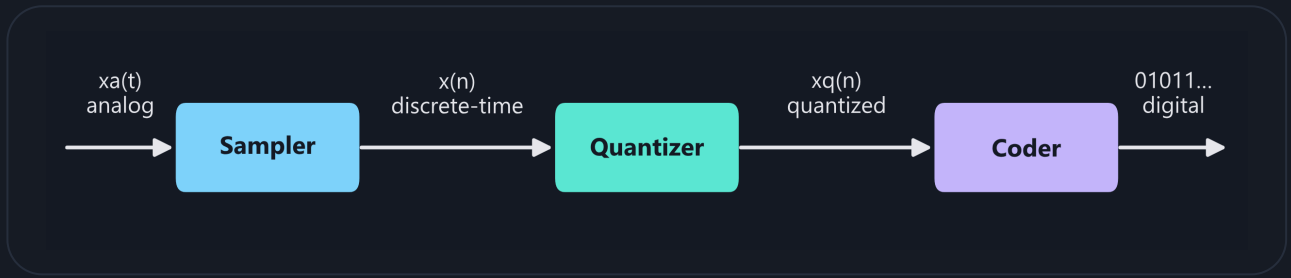
Lecture **LEC 4** · Sec. 1.6.1 · p. 8

Came in Final 20-1-2025 · Q4-a | Final 27-6-2021 · Q1-a | Midterm 26-7-2026 · Q2-a | Midterm 31-7-2023 · Q2-a | Midterm 2 · Q1-a | Quiz 1 Summer 26 · Q1-a

Q13 ● Frequent 4×

Explain (with a block diagram) the basic three operations performed in the A/D converter.

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

Key idea الـ A/D تلات خطوات ورا بعض: الـ sampler بيقطّع الوقت، والـ quantizer بيقطّع القيم، والـ coder بيحوّل كل level لـ binary.

Key words Sampling Quantization Coding

Lecture 4

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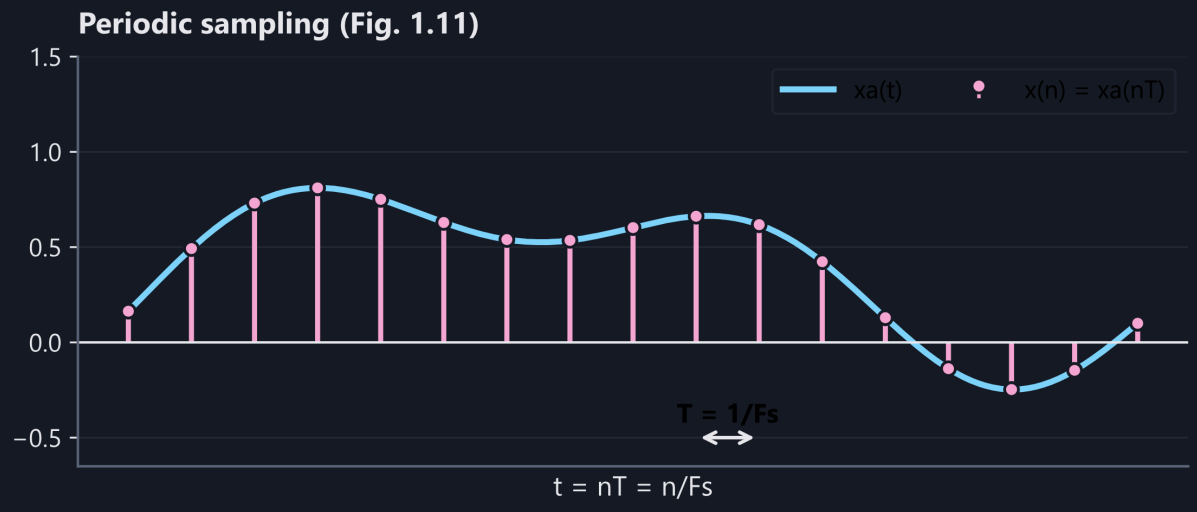
Lecture **LEC 4** · Sec. 1.6 · p. 3

Came in Revision Sheet 1 | Midterm OLD · Model-A 2012/2013 | Sheet 1 · 1.4 | Midterm theory notes

Q14 ● Frequent 4×

In A/D, explain in detail the sampling process of analog signals.

- We use periodic (uniform) sampling, described by the relation **x(n) = xa(nT)**, $-\infty < n < \infty$ (1.29), where x(n) is obtained by taking samples of xa(t) every T seconds (Fig. 1.11).
- The time interval T between successive samples is the **sampling period**; its reciprocal **Fs = 1/T** is the sampling rate (samples/s) or sampling frequency (Hz).
- Periodic sampling relates the time variables t and n: **t = nT = n/Fs** (1.30). As a consequence, the frequencies are related by F = f/Fs (1.33).
- Sampling does not result in a loss of information if the bandwidth is finite and Fs > $2f_{max}$.



Key idea بناخد sample كل T ثانية: x(n) = xa(nT)، و T = 1/Fs.

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Lecture **LEC 4** · Sec. 1.6.1 · p. 8

Came in Midterm 28-11-2021 · Q1-a | Midterm 8-12-2020 · Q1-a | Midterm 1 Model-2 · Q1-a | Midterm OLD

Q22 ● Frequent 4×

Check (✓) for the right statements and (×) for the wrong statements and correct the mistakes.

1. In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. **(×)**
   Correct: when Δ **decreases**, the accuracy increases.
2. Nyquist frequency (F_N) is the minimum sampling frequency. **(✓)**
3. To reconstruct x(t) from x(n), the sampling rate Fs must be less than the frequency of x(t), f_o. **(×)**
   Correct: Fs must be **greater than $2f_o$**.

Key idea Δ أصغر معناها دقة أعلى، فالجملة الأولى غلط. الـ Nyquist rate هو أقل sampling rate، فدي صح. وعشان نرجّع الإشارة لازم Fs > 2f0، فالأخيرة غلط.

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6 · p. 3

Came in Final 12-1-2019 · Q5 | Final 23-12-2017 · Q5-b | Final 27-6-2021 · Q4 | Quiz 1 Summer 26 · Q3

Q17 3×

State and explain the sampling theorem. Define the Nyquist rate.

- To select Fs we must have information about the frequency content of the signal, e.g. f_{max} = 3000 Hz for speech and 5 MHz for television signals. The signal may be passed through a filter that severely attenuates components above f_{max}.
- To avoid aliasing, Fs is selected so that **$Fs > 2f_{max}$** (1.47). Then every component $|f_i| < f_{max}$ is mapped into $-1/2 \le F_i = f_i/Fs \le 1/2$ (1.48).

Sampling theorem: if the highest frequency contained in an analog signal xa(t) is $f_{max} = B$ and the signal is sampled at a rate $Fs > 2f_{max} = 2B$, then xa(t) can be exactly recovered from its sample values using the interpolation function:

```
g(t) = sin(2πBt) / (2πBt) (1.50)     xa(t) = Σ xa(n/Fs) g(t − n/Fs) (1.51)
```

- The sampling rate **$f_N = 2B = 2f_{max}$** is called the **Nyquist rate**.
- Because of the complexity and the infinite number of samples required, these reconstruction formulas are primarily of theoretical interest.

Key idea لو الإشارة band-limited وأعلى تردد فيها fmax، نقدر نرجّعها بالظبط من الـ samples لو Fs > 2fmax، والـ Nyquist rate هو 2fmax.

Key words Sampling theorem

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Sheet 1 · 1.5 · Midterm theory notes · Eng. Khalaf theory notes

Q19 1×

In A/D conversion, explain in detail the coding process.

- The coding process assigns a unique binary number to each quantization level.
- With L levels we need at least L binary numbers; with b bits we can create 2^b numbers.

$$2^b \geq L \rightarrow b \geq \log_2 L$$

Key idea كل level بياخد كود binary مختلف من b bits، ولازم $b \geq L^2$.

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.6.4 · p. 32

Came in Book exam Midterm 2015 · Model-A

Q20 Less frequent

Why is the sampling process invertible while the quantization process is not?

- **Sampling** does not result in a loss of information, nor introduce distortion, if the signal bandwidth is finite → **invertible (reversible)**.
- **Quantization** is a **noninvertible (irreversible)** process that results in signal distortion.

Key idea الـ sampling ما بيضيّعش معلومات لو الإشارة band-limited، فنقدر نرجّعها. الـ quantization بيرمي القيمة الأصلية، فالـ error ده ما يرجعش.

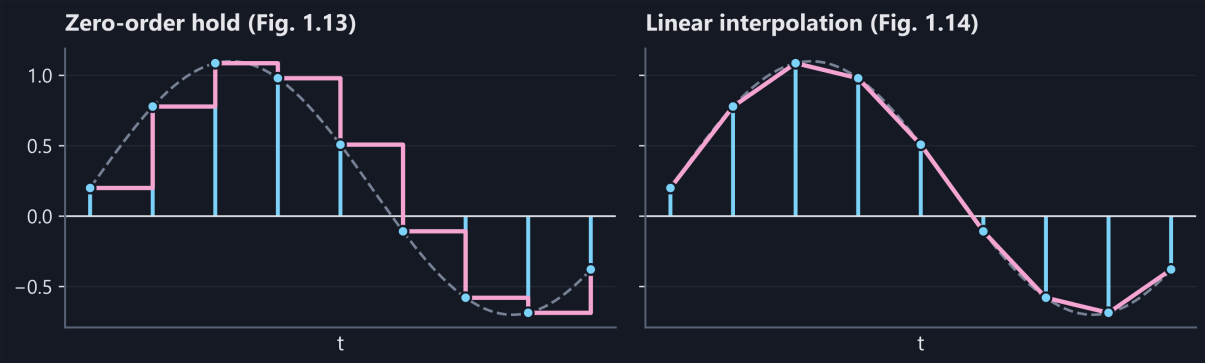
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Lecture **LEC 4** · Sec. 1.6.3 · p. 23

Q21 Less frequent

Explain the D/A conversion process. Compare between ideal interpolation, zero-order hold and linear interpolation.

- The task of a D/A converter is to **interpolate between samples**. The ideal interpolation (sampling theorem) is too complicated and impractical.
- **Zero-order hold:** the simplest; holds constant the value of one sample until the next one is received.
- **Linear interpolation:** connects successive samples with straight-line segments (better).



Key idea الـ D/A بيوصّل الـ samples ببعض. الـ ideal interpolation صعب عمليًا، والـ zero-order hold بيثبّت كل قيمة لحد اللي بعدها، والـ linear interpolation بيوصّلهم بخطوط مستقيمة.

Key words Zero-order hold | Linear interpolation

Source — from your PDFs

Lecture **LEC 4** · Sec. 1.7 · p. 34

★ Cheat sheet · Lecture 4

Remember

Q13 A/D = sampler (discrete time) → quantizer (discrete values) → coder (b-bit binary words).

Q14 Periodic sampling takes x(n) = xa(nT), one sample every T = 1/Fs seconds.

Q15 t = nT gives F = f / Fs (and ω = Ω T): the digital frequency is the analog frequency divided by the sampling rate.

Q16 $|F| \le 1/2$ with F = f/Fs → $f \le Fs/2$ for every component → Fs ≥ 2 fmax; otherwise higher frequencies alias.

Q17 A band-limited signal (fmax) is recovered exactly from its samples if Fs > 2 fmax; FN = 2 fmax is the Nyquist rate.

Q18 Each sample is mapped to one of L levels with step $\Delta = (x_{max} - x_{min})/(L - 1)$; the error eq(n) is at most Δ/2 when rounding.

Q19 Each quantized level gets a unique b-bit binary number, with $2^b \ge L$.

Q20 Sampling a band-limited signal loses nothing (invertible); quantization throws away the exact value, so the error cannot be undone.

Q21 D/A connects the samples: ideal interpolation is impractical, zero-order hold keeps each value constant, linear interpolation joins samples with straight lines.

Q22 Smaller Δ → more accurate (× as written); Nyquist rate = minimum sampling rate (✓); reconstruction needs Fs > $2f_o$ (× as written).

Equations

$$x(n) = x_a(nT) = A\cos(2\pi f nT + \theta) = A\cos(2\pi n f/Fs + \theta) \quad (1.32)$$

$$F = f / Fs \quad (1.33) \qquad \text{or} \qquad \Omega = \omega T \quad (1.34)$$

$$-1/2 \le f/Fs \le 1/2 \rightarrow -Fs/2 \le f \le Fs/2 \quad (1.37)$$

$$f_{max} = Fs/2 \rightarrow Fs \ge 2f_{max}$$

$$g(t) = \sin(2\pi Bt) / (2\pi Bt) \quad (1.50)$$

$$x_a(t) = \Sigma\, x_a(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

SECTION 04

Solving Method

The doctor's steps for every problem type

Questions

- ☐ M1 x(t) = A cos(2πft + θ) and Fs: reconstruct? x(n)? Np? samples? plot p. 28
- ☐ M2 Several components / low Fs: x(n), aliasing and the reconstructed ya(t) p. 29
- ☐ M3 Is it periodic? Fundamental period p. 29
- ☐ M4 Nyquist rate, alias frequency and quantization p. 30

M1

x(t) = A cos(2πft + θ) and Fs: reconstruct? x(n)? Np? samples? plot

1. Compare with the standard form: **x(t) = A cos(2πft + θ)** → write A (volt), θ (rad) and f (Hz).

2. $f_{max} = f$ → **$Fs \geq 2f_{max}$**. If the given Fs satisfies it: "Yes, we can reconstruct the original signal successfully from its samples" (no aliasing). If not: aliasing occurs.

3. Sample: **t → nTs = n/Fs** ⇒ x(n) = xa(nTs) = A cos(2πf n/Fs + θ). Simplify the angle.

4. Fundamental period: **2πFn = angle** → F = f/Fs = k/N_P (k, N_P relatively prime) → **N_P** samples/cycle.

5. Sample values for **n = 0 … N_P − 1**: x(0), x(1), …

6. Plot x(t) and x(n) on the same figure: the continuous curve plus stems with dots at the samples.

DOCTOR'S STYLE

Examples: Midterm 14-11-2019 (Q1-c) solved by the doctor · P8 to P12 · Sheet 1.7

Solving Method

M2

Several components / low Fs: x(n), aliasing and the reconstructed ya(t)

1. List the frequencies present: f1, f2, f3 … → f_{max} → Nyquist rate **$f_N = 2f_{max}$**.

2. Folding frequency **Fs/2**: the maximum frequency that can be represented uniquely by the sampled signal.

3. x(n) = xa(nT) = xa(n/Fs): write every term as cos 2π(**f/Fs**)n.

4. If f/Fs > 1/2, subtract an integer to bring it into [−1/2, 1/2]: e.g. 2π(**6/5**)n = 2π(1 + 1/5)n → 2π(1/5)n, and sin 2π(3/5)n = sin 2π(1 − 2/5)n = −sin 2π(2/5)n. Collect terms with the same frequency.

5. ya(t): replace n by **Fs·t** (ideal interpolation). For a different D/A use n = t/T2. Say that ya(t) differs from xa(t) because of aliasing.

DOCTOR'S STYLE
Examples: Example 1.3 (lecture) · Sheet 1.8 (T1 = 5 ms, T2 = 1 ms)

Key words Fs/2

M3

Is it periodic? Fundamental period

1. Continuous-time sinusoid: **always periodic**, 2πf = ω → **Tp = 1/f**. Show xa(t + Tp) = xa(t).

2. Discrete-time sinusoid: periodic **only if F is rational**. 2πF = Ω → F = k/N (cancel common factors → k, N relatively prime) → **N = fundamental period**. Show x(n + N) = x(n).

3. If F contains π (e.g. Ω = 2 → F = 1/π) it is not a ratio of integers → **not periodic**.

DOCTOR'S STYLE
Examples: Lec 3 (F = 31/60, Ω_o = π/8 …) · Sheet 1.6 · P7

M4

Nyquist rate, alias frequency and quantization

1. Nyquist rate: $\mathbf{f_N = 2f_{max}}$; the sampling theorem: Fs > $2f_{max}$ (sampling higher than the Nyquist rate is preferred, Example 1.2).

2. Alias: at sampling rate Fs, $\mathbf{f_k = f_0 + kFs}$ give identical samples. Find the alias in 0 < f < Fs/2 from F = f/Fs (Example 1.1-d, Example 1.4).

3. Quantization: $\mathbf{\Delta = (x_{max} - x_{min})/(L - 1)}$, rounding error $|e_q(n)| \le \Delta/2$, and b bits with $2^b \ge L$ (Table 1.2).

DOCTOR'S STYLE

Examples: Examples 1.1, 1.2 and 1.4 · P5 (Table 1.2)

SECTION 05

Lecture Examples

Solved step by step

Questions

- ☐ P1 Example 1.1 — Sampling & aliasing p. 31
- ☐ P2 Example 1.2 — Nyquist rate p. 33
- ☐ P3 Example 1.3 — Aliasing with three components p. 35
- ☐ P4 Example 1.4 — The aliasing concept p. 36
- ☐ P5 Quantization example (Table 1.2) p. 38
- ☐ P6 Periodicity examples (Lec 3) p. 40

P1 3×

Example 1.1 — Sampling & aliasing

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$.

a) Determine the minimum sampling rate required to avoid aliasing.

b) The signal is sampled at Fs = 200 Hz. What is the discrete-time signal?

c) The signal is sampled at Fs = 75 Hz. What is the discrete-time signal?

d) What is the frequency $0 < f < Fs/2$ of a sinusoid that yields samples identical to those in (c)?

a) The frequency of the analog signal is f = 50 Hz. Hence the minimum sampling rate required to avoid aliasing is Fs = 100 Hz.

b) If the signal is sampled at Fs = 200 Hz, the discrete-time signal is

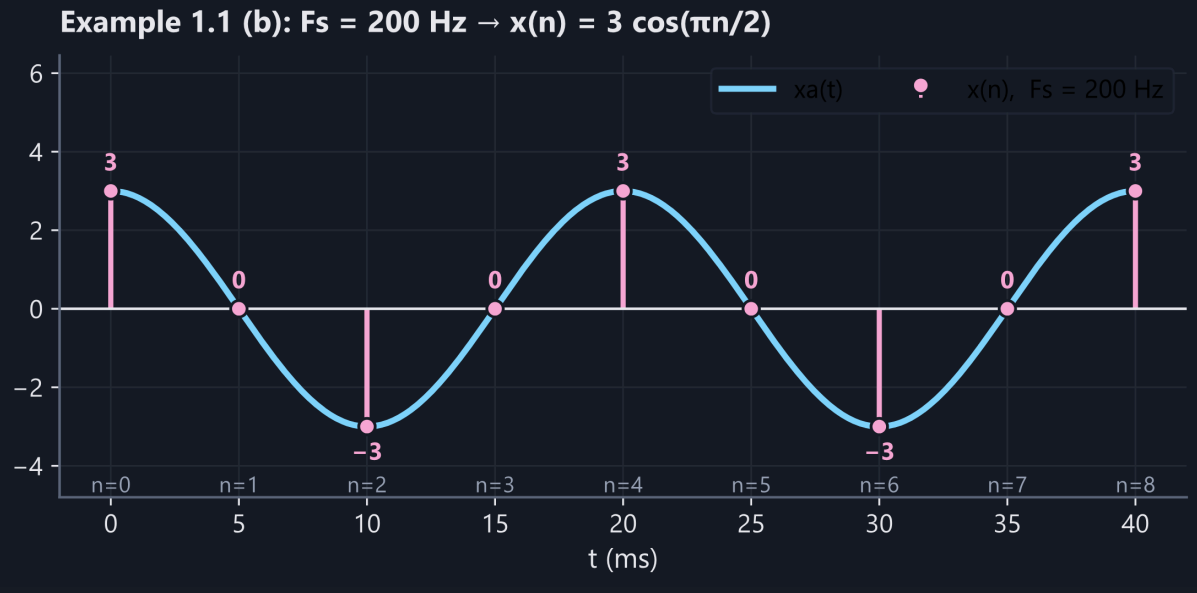
$x(n) = 3 \cos(100\pi/200)n = 3 \cos(\pi/2)n$

c) If the signal is sampled at Fs = 75 Hz, the discrete-time signal is

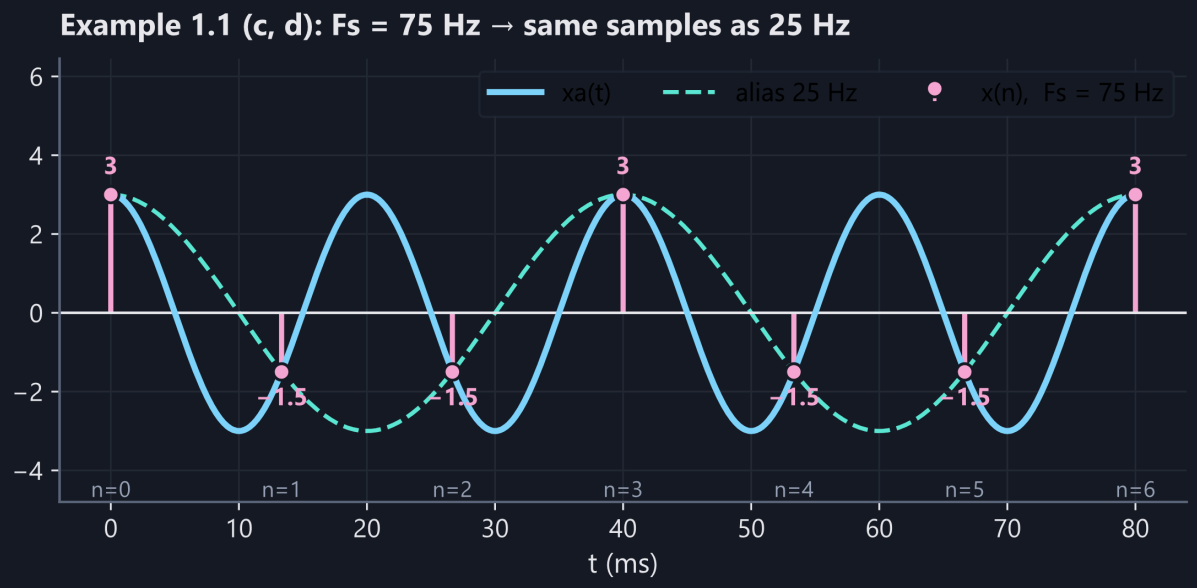
$x(n) = 3 \cos(100\pi/75)n = 3 \cos(4\pi/3)n = 3 \cos(2\pi - 2\pi/3)n = 3 \cos(2\pi/3)n$

d) For the sampling rate of Fs = 75 Hz, we have f = F·Fs = 75F. The frequency of the sinusoid in part (c) is F = 1/3. Hence f = 25 Hz.

Clearly, the sinusoidal signal ya(t) = 3 cos 2πft = 3 cos 50πt sampled at Fs = 75 samples/s yields identical samples. Hence f = 50 Hz is an alias of f = 25 Hz for the sampling rate Fs = 75 Hz.



Lecture Examples



Result Fs,min = 100 Hz · Fs = 200 Hz → x(n) = 3 cos(πn/2) · Fs = 75 Hz → x(n) = 3 cos(2πn/3) · 25 Hz gives the same samples (50 Hz is its alias)

Key idea نطلع f من المعادلة ونقارن Fs بـ 2f. لو Fs أقل يحصل aliasing، والتردد اللي بيظهر هو اللي بيدّي نفس الـ samples جوه المدى من 0 لحد Fs/2.

Source — from your PDFs

Lecture **LEC 4** · Example 1.1 · p. 38

Came in Final 20-1-2025 · Final 11-6-2023 · Final 23-1-2023 (same type: is Fs enough? + x(n))

Lecture Examples

P2 3×

Example 1.2 — Nyquist rate

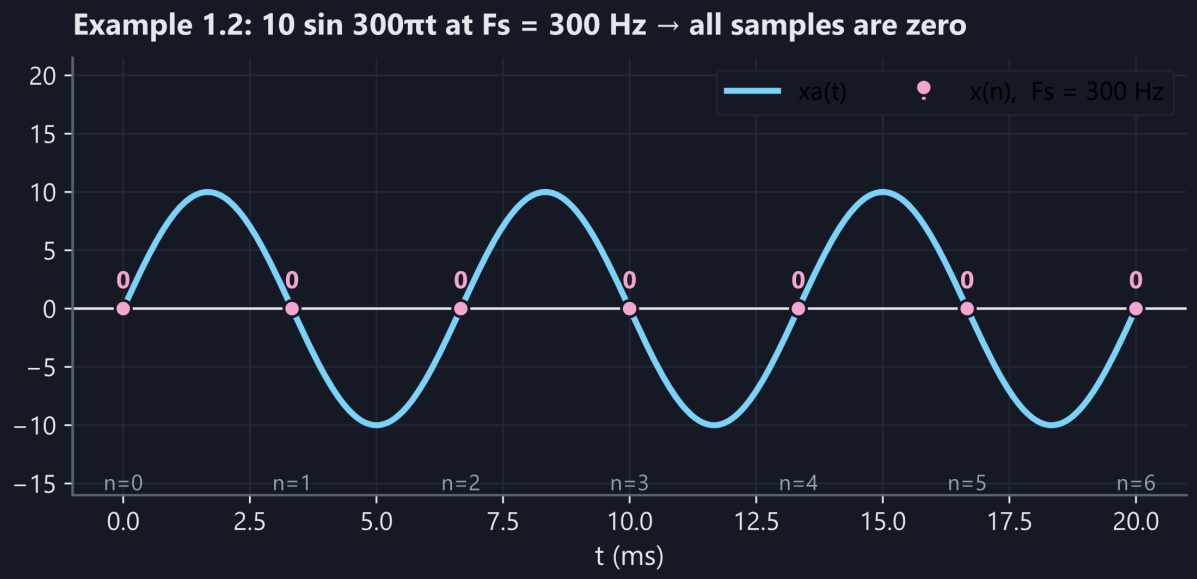
Consider $x_a(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$. What is the Nyquist rate for this signal?

The frequencies present in the signal above are f1 = 25 Hz, f2 = 150 Hz, f3 = 50 Hz. Thus fmax = 150 Hz and according to (1.47), Fs > 2fmax = 300 Hz. The Nyquist rate is fN = 2fmax. Hence fN = 300 Hz.

Discussion

- It should be observed that the signal component 10 sin 300πt, sampled at the Nyquist rate fN = 300, results in the samples 10 sin πn, which are identically zero.
- In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely.
- This situation would not occur if the sinusoid is offset in phase by some amount θ. In such a case we have 10 sin(300πt + θ) sampled at the Nyquist rate fN = 300 samples per second, which yields the samples:
  10 sin(πn + θ) = 10(sin πn cos θ + cos πn sin θ) = 10 sin θ cos πn = $(-1)^n$ 10 sin θ
- Thus if θ ≠ 0 or π, the samples of the sinusoid taken at the Nyquist rate are not all zero.
- However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown.
- A simple remedy that avoids this potentially troublesome situation is to sample the analog signal at a rate higher than the Nyquist rate.

Lecture Examples



Result fN = 300 Hz; sampling exactly at fN can miss the 150 Hz component (samples = $(-1)^n$ 10 sin θ) → sample above the Nyquist rate

Key idea نطلع كل الترددات اللي في الإشارة ونختار أكبرها fmax، والـ Nyquist rate يساوي 2fmax.

Source — from your PDFs

Lecture **LEC 4** · Example 1.2 · p. 41

Came in Final 12-1-2019 · Final 28-12-2019

P3

Example 1.3 — Aliasing with three components

Consider $x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12000\pi t$.

a) What is the Nyquist rate?

b) The signal is sampled at Fs = 5000 samples/s. What is the discrete-time signal?

c) What is the analog signal $y_a(t)$ reconstructed using ideal interpolation?

(a) The frequencies existing in the analog signal are f1 = 1 kHz, f2 = 3 kHz, f3 = 6 kHz. Thus fmax = 6 kHz, and according to the sampling theorem, Fs > 2fmax = 12 kHz. The Nyquist rate is fN = 12 kHz.

(b) Since we have chosen Fs = 5 kHz, the folding frequency is Fs/2 = 2.5 kHz, and this is the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.30) we obtain:

$x(n) = xa(nT) = xa(n/Fs)$
$= 3\cos 2\pi(1/5)n + 5\sin 2\pi(3/5)n + 10\cos 2\pi(6/5)n$
$= 3\cos 2\pi(1/5)n + 5\sin 2\pi(1 - 2/5)n + 10\cos 2\pi(1 + 1/5)n$
$= 3\cos 2\pi(1/5)n + 5\sin 2\pi(-2/5)n + 10\cos 2\pi(1/5)n$

Finally, we obtain:

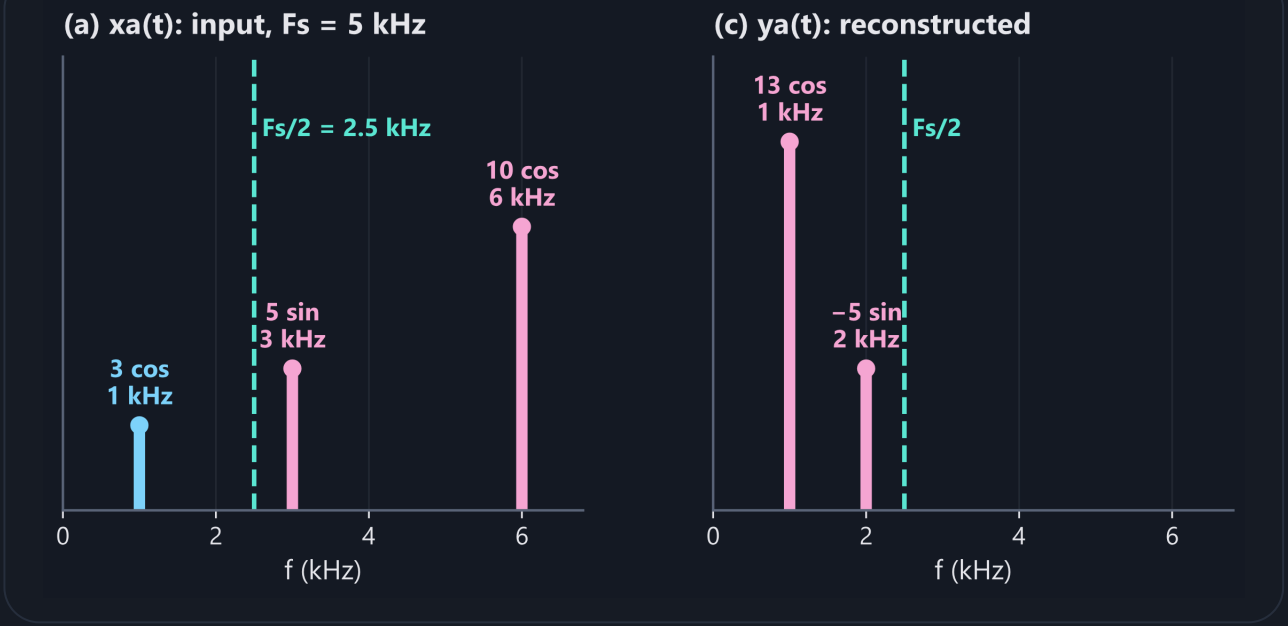
$x(n) = 13\cos 2\pi(1/5)n - 5\sin 2\pi(2/5)n$

(c) Since only the frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analog signal we can recover is

$ya(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$

which is obviously different from the original signal xa(t). This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

Lecture Examples



Result $f_N = 12$ kHz · $x(n) = 13\cos 2\pi(1/5)n - 5\sin 2\pi(2/5)n$ · $y_a(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$ (aliasing)

Key idea الـ Fs أقل من 2fmax، فالترددات اللي فوق Fs/2 بتتطوي لترددات أقل وبتتجمع مع اللي موجود، فالإشارة اللي بنرجّعها بتطلع مختلفة عن الأصلية.

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Lecture **LEC 4** · Example 1.3 · p. 44

P4

Example 1.4 — The aliasing concept

Two signals x1(t) = cos 2π(10)t and x2(t) = cos 2π(50)t are sampled at Fs = 40 Hz. Find x1(n) and x2(n) and discuss.

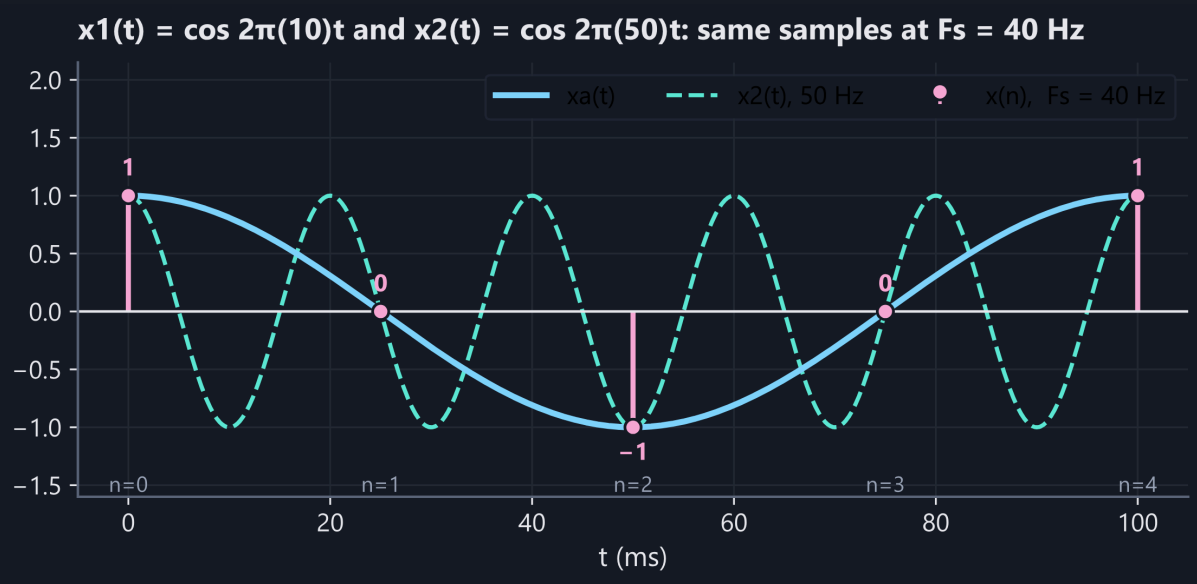
The implications of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals x1(t) = cos 2π(10)t and x2(t) = cos 2π(50)t, which are sampled at a rate Fs = 40 Hz. The corresponding discrete-time signals or sequences are:

x1(n) = cos 2π(10/40)n = cos (π/2)n

x2(n) = cos 2π(50/40)n = cos (5π/2)n

Lecture Examples

- However, cos 5πn/2 = cos(2πn + πn/2) = cos πn/2. Hence x2(n) = x1(n).
- Thus the sinusoidal signals are identical and, consequently, indistinguishable.
- If we are given the sampled values generated by cos(π/2)n, there is some ambiguity as to whether these sampled values correspond to x1(t) or x2(t).
- Since x2(t) yields exactly the same values as x1(t) when the two are sampled at Fs = 40 samples per second, we say that the frequency f2 = 50 Hz is an alias of the frequency f1 = 10 Hz at the sampling rate of 40 samples per second.
- It is important to note that f2 is not the only alias of f1. In fact at the sampling rate of 40 samples per second, the frequency f3 = 90 Hz is also an alias of f1, as is the frequency f4 = 130 Hz, and so on.



Result At Fs = 40 Hz, f2 = 50 Hz is an alias of f1 = 10 Hz (so are 90 Hz, 130 Hz, …)

Key idea إشارتين بترددين مختلفين بيدّوا نفس الـ samples بالظبط عند Fs = 40 Hz، فما نقدرش نفرّق بينهم. ده معنى الـ alias.

Source — from your PDFs

Lecture **LEC 4** · Example 1.4 · p. 48

P5

Quantization example (Table 1.2)

$x_a(t) = 0.9^t$, $t \geq 0$ is sampled at Fs = 1 Hz, giving $x(n) = 0.9^n$. Quantize the first 10 samples to one significant digit by truncation and by rounding; find the error, Δ, and the number of bits.

- Let us consider the discrete-time signal $x(n) = 0.9^n$ for $n \geq 0$ (0 for $n < 0$), obtained by sampling the analog exponential signal $xa(t) = 0.9^t$, $t \geq 0$ with a sampling frequency Fs = 1 Hz (Ts = 1 sec), see Fig. 1.15(a).
- Observation of Table 1.2, which shows the values of the first 10 samples of x(n), reveals that the description of the sample value x(n) requires n significant digits.
- However, let us assume that we want to use only one significant digit. To eliminate the excess digits, we can either simply discard them (truncation) or discard them by rounding the resulting number (rounding).
- The resulting quantized signals xq(n) are shown in Table 1.2. We discuss only quantization by rounding, although it is just as easy to treat truncation. The rounding process is graphically illustrated in Fig. 1.15b.
- In our example we have xmax = 1, xmin = 0, and L = 11, which leads to Δ = 0.1.
- In our example it can easily be seen that we need a coder with b = 4 bits.

Lecture Examples

n	x(n)	Truncation	Rounding	eq(n)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	−0.01
3	0.729	0.7	0.7	−0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	−0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	−0.03046721
9	0.387420489	0.3	0.4	0.012579511

$$\Delta = (x_{max} - x_{min})/(L - 1) = (1 - 0)/(11 - 1) = \mathbf{0.1} \qquad e_q(n) = x_q(n) - x(n),$$

$$|e_q(n)| \le \Delta/2 = 0.05$$

$$2^b \ge L = 11 \rightarrow \mathbf{b = 4\ bits}$$



Lecture Examples

Result $\Delta = 0.1,\ L = 11,\ |eq(n)| \le \Delta/2 = 0.05,\ b = 4$ bits

Key idea نحسب Δ من مدى الإشارة وعدد الـ levels، ونقرّب كل sample مرة بالـ truncation ومرة بالـ rounding، ونقارن الـ error في الحالتين.

Source — from your PDFs

Lecture **LEC 4** · Table 1.2 · p. 27

P6

Periodicity examples (Lec 3)

a) Find the fundamental period of a DT sinusoid with F1 = 31/60 and F2 = 30/60.

b) For x(n) = cos $\Omega_o n$, find N for Ω_o = 0, π/8, π/4, π/2, π.

c) Fig. 1.7: Ω = π/6. Find F and N.

F = k/N (1.16): cancel common factors so that k and N are relatively prime → N is the fundamental period

a) F1 = 31/60 → k = 31, N = 60 (relatively prime) → **N1 = 60**. F2 = 30/60 = 1/2 → **N2 = 2**. A small change in frequency results in a large change in the period.

b) $x(n) = \cos \Omega_0 n$, $F = \Omega_0/2\pi$:

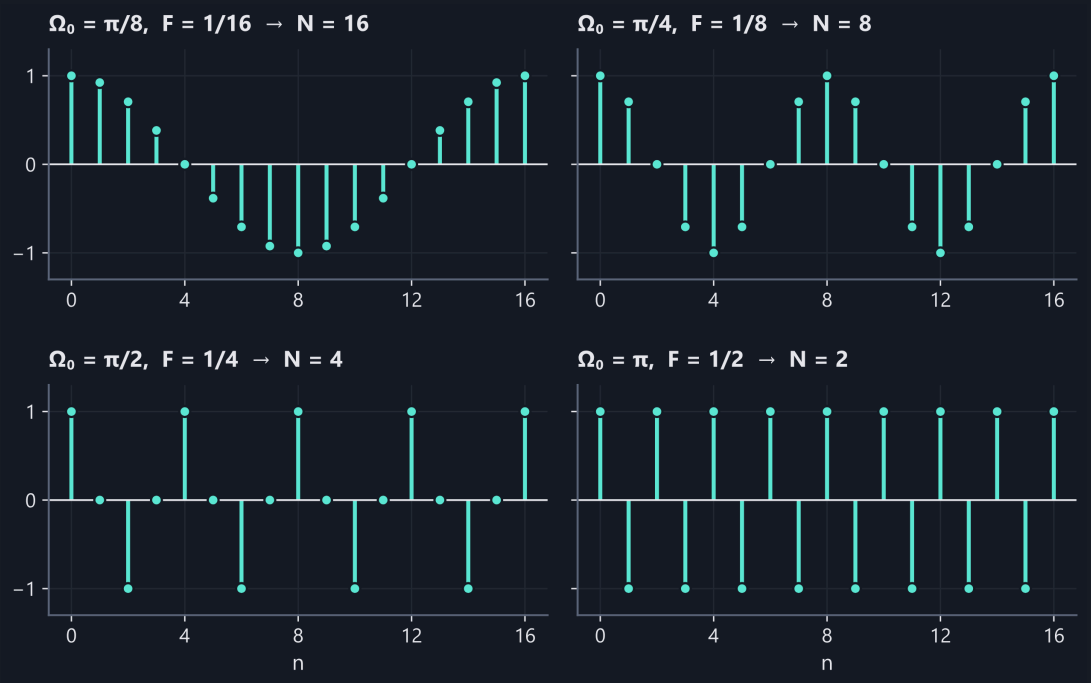
$\Omega_0 = 0 \to F = 0 \to N = \infty$ | $\pi/8 \to 1/16 \to N = 16$ | $\pi/4 \to 1/8 \to N = 8$

$\pi/2 \to 1/4 \to N = 4$ | $\pi \to 1/2 \to N = 2$

The period decreases as the frequency increases (highest rate of oscillation at Ω = π).

c) $\Omega = \pi/6 \to F = (\pi/6)/2\pi = 1/12 \to$ **N = 12**

Lecture Examples



Result a) N1 = 60, N2 = 2 · b) Ω_0 = 0, π/8, π/4, π/2, π → N = ∞, 16, 8, 4, 2 · c) N = 12

Key idea نكتب F = k/N ونختصر الكسر، والمقام بعد الاختصار هو الـ period. تغيير صغير في F ممكن يغيّر الـ period جامد.

Source — from your PDFs

Lecture **LEC 3** · Sec. 1.5.2 · p. 9

★ Cheat sheet · Lecture Examples

Final results

P1 Fs,min = 100 Hz · Fs = 200 Hz → x(n) = 3 cos(πn/2) · Fs = 75 Hz → x(n) = 3 cos(2πn/3) · 25 Hz gives the same samples (50 Hz is its alias)

P2 fN = 300 Hz; sampling exactly at fN can miss the 150 Hz component (samples = $(-1)^n$ 10 sin θ) → sample above the Nyquist rate

P3	fN = 12 kHz · x(n) = 13 cos 2π(1/5)n − 5 sin 2π(2/5)n · ya(t) = 13 cos 2000πt − 5 sin 4000πt (aliasing)
P4	At Fs = 40 Hz, f2 = 50 Hz is an alias of f1 = 10 Hz (so are 90 Hz, 130 Hz, ...)
P5	Δ = 0.1, L = 11, $\lvert eq(n)\rvert \le \Delta/2 = 0.05$, b = 4 bits
P6	a) N1 = 60, N2 = 2 · b) Ω_o = 0, π/8, π/4, π/2, π → N = ∞, 16, 8, 4, 2 · c) N = 12

SECTION 06

Previous Exams

Midterms, finals, quizzes & Revision Sheet

Questions

☐	P7	Periodicity of continuous-time and discrete-time signals	p. 43
☐	P8	x(t) = cos(200πt), Fs = 400 Hz	p. 45
☐	P9	x(t) = 10 cos(1500πt), Fs = 3000 samples/s	p. 47
☐	P10	x(t) = 10 cos(1000πt), Fs = 2000 sample/s	p. 49
☐	P11	x(t) = A cos(2πf$_o$t), A = 2 V, f$_o$ = 10 kHz	p. 51
☐	P12	x(t) = 2 cos(2000πt), Fs = 3000 Hz then Fs = 8000 Hz	p. 53
☐	P14	Is Fs suitable? (Fs = Nyquist rate exactly)	p. 54
☐	P15	Two signals with two sampling frequencies	p. 56
☐	P17	x(n) = 2 cos(nπ/4): periodicity and first four samples	p. 58

P7 Frequent 9×

Periodicity of continuous-time and discrete-time signals

Check the periodicity of the following signals and find their fundamental periods:
(i) $x_a(t) = 3\cos(5t + \pi/6)$ (ii) $x(n) = \cos(3\pi n)$ (iii) $x(n) = \cos(0.01\pi n)$ (iv) $X(n) = 2\sin(3\pi n/105)$ (v) $x(t) = \cos(3000\pi t)$ (vi) $y(t) = 7\cos(400\pi t)$ (vii) $x(n) = \sin(2n + \pi/4)$

(i)

$x_a(t) = 3\cos(5t + \pi/6)$ is a continuous-time sinusoid $x_a(t) = A\cos(2\pi ft + \theta)$. For every fixed value of f, $x_a(t)$ is periodic with $T_p = 1/f$.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{T_p = 1/f = 2\pi/5 \text{ sec}}$$
$$x_a(t + T_p) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

$\therefore$ Periodic, fundamental period $T_p = 2\pi/5$ sec.

(ii)

$x(n) = \cos(3\pi n)$ is a discrete-time sinusoid $x(n) = A\cos(2\pi Fn + \theta)$: it is periodic only if F is rational ($F = k/N$).

$$2\pi F = 3\pi \rightarrow F = 3/2 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 2) = \cos(3\pi n + 6\pi) = \cos(3\pi n) = x(n)$$

$\therefore$ Periodic, fundamental period **N = 2 samples**.

(iii)

$x(n) = \cos(0.01\pi n)$ is a discrete-time sinusoid $x(n) = A\cos(2\pi Fn + \theta)$: it is periodic only if F is rational ($F = k/N$).

$$2\pi F = 0.01\pi \rightarrow F = 1/200 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 200) = \cos(0.01\pi n + 2\pi) = x(n)$$

$\therefore$ Periodic, fundamental period **N = 200 samples**.

(iv)

Previous Exams

x(n) = 2 sin(3πn/105) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi/105 = \pi/35 \rightarrow F = 1/70 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 70) = 2\sin(\pi n/35 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 70 samples**.

(v)

x(t) = cos(3000πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 3000\pi \rightarrow f = 1500 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 1/1500 \text{ sec} = 0.667 \text{ msec}}$$
$$x(t + Tp) = \cos(3000\pi t + 2\pi) = x(t)$$

∴ Periodic, fundamental period Tp = 1/1500 sec = 0.667 msec.

(vi)

y(t) = 7 cos(400πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 400\pi \rightarrow f = 200 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 5 \text{ msec}}$$
$$y(t + Tp) = 7\cos(400\pi t + 2\pi) = y(t)$$

∴ Periodic, fundamental period Tp = 5 msec.

(vii)

x(n) = sin(2n + π/4) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 2 \rightarrow F = 2/2\pi = 1/\pi$$

F cannot be expressed as a ratio of two integers → **not periodic**.

Result Tp = 2π/5 s · N = 2 · N = 200 · N = 70 · Tp = 1/1500 s · Tp = 5 ms · sin(2n + π/4) not periodic

Key idea الـ continuous دايمًا periodic و Tp = 1/f. الـ discrete لازم F تبقى rational، ولو F طلع فيها π تبقى الإشارة مش periodic.

Source — from your PDFs

Lecture	**LEC 3** · Sec. 1.5.1 · p. 3
Lecture	**LEC 3** · Sec. 1.5.2 · p. 9
Came in	Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm 1 Model-2 · Midterm 8-12-2020 · Final 27-6-2021 · Final 12-1-2019 · Final 23-12-2017 · Final 2-1-2017 · Midterm OLD

P8 · Frequent 8×

x(t) = cos(200πt), Fs = 400 Hz

For the signal x(t) = cos(200πt) to be sampled with Fs = 400 Hz:

(i) Find an expression for its corresponding sequence x(n). Then calculate its fundamental period N_P.

(ii) Find the sample values of the first cycle in x(n), then plot x(n) and x(t) on the same figure.

$$x(t) = 1\cos(200\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 1, \quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100\ Hz}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200\ Hz}$$
$$Fs = 400\ Hz \geq 2f_{max} = 200\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(i)

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

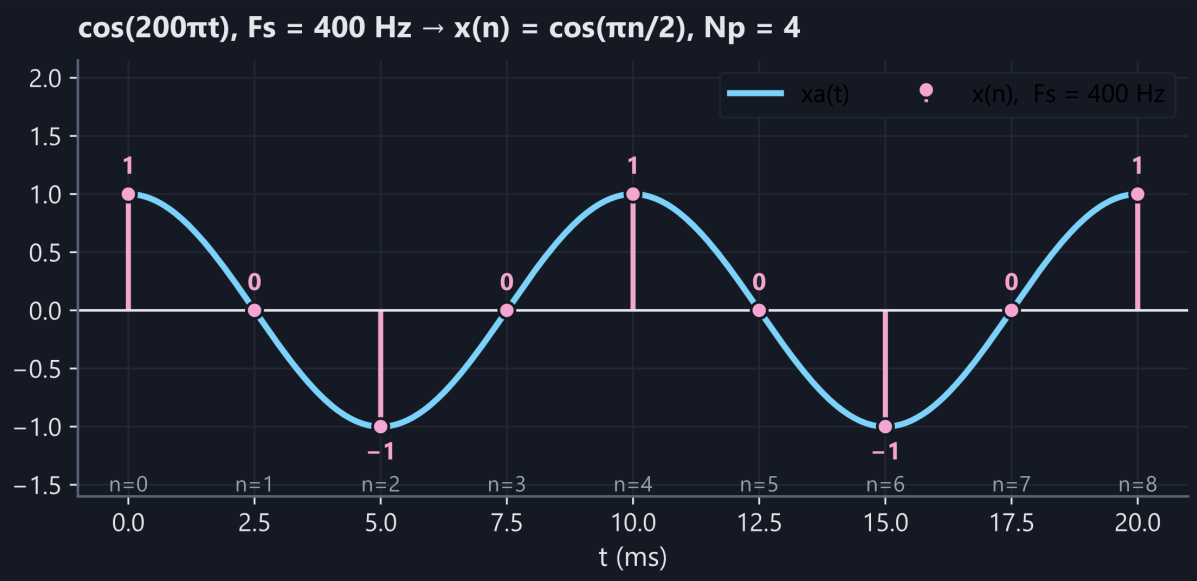
$$x(n) = x_a(nT) = x_a(n/Fs) = 1\cos(200\pi\, n/400) = \mathbf{1\cos(\pi/2\, n)}$$

$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 1\cos(0) = \mathbf{1}$$
$$x(1) = 1\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 1\cos(\pi) = \mathbf{-1}$$
$$x(3) = 1\cos(3\pi/2) = \mathbf{0}$$

(ii) Plot x(t) and x(n) on the same figure:



Result fN = 200 Hz, no aliasing · x(n) = cos(πn/2), Np = 4: 1, 0, −1, 0

Key idea نوع M1: f = 100 Hz و Fs = 400 أكبر من 2f، فمفيش aliasing. نعوّض t = n/Fs فنطلع F = 1/4 يعني Np = 4، ونحسب أربع samples.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Midterm 26-7-2026 · Midterm 31-7-2023 · Midterm 1 Model-2 · Final 7-6-2023 · Q4-b · Final 8-9-2021 · Final 18-1-2024 · Quiz 1 · Section · Midterm OLD · several models

P9 3×

x(t) = 10 cos(1500πt), Fs = 3000 samples/s

For x(t) = 10 cos(1500πt):
(i) If it is sampled with Fs = 3000 sample/s, is it possible to reconstruct this signal from its samples? Find the Nyquist frequency F_N.
(ii) Using Fs, find x(n), then calculate the sample values during one cycle and plot x(t) and x(n).

$$x(t) = 10\cos(1500\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1500\pi \rightarrow \mathbf{f = 750\ Hz}$$

(i)

$$F_N = 2f_{max} = 2 \times 750 = \mathbf{1500\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 1500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(ii)

If the signal is sampled at Fs = 3000 Hz (t = nT = n/Fs), the discrete-time signal is:

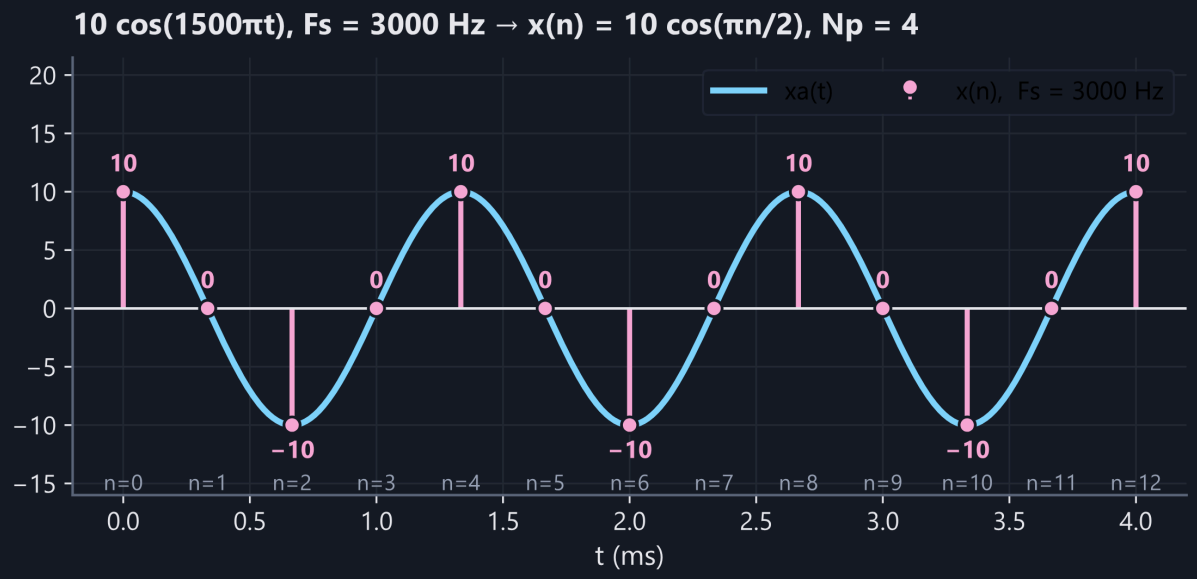
$$x(n) = xa(nT) = xa(n/Fs) = 10\cos(1500\pi\ n/3000) = \mathbf{10\cos(\pi/2\ n)}$$

$$F = f/Fs = 750/3000 = \mathbf{1/4} = k/N_p \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_p = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 10\cos(\pi) = \mathbf{-10}$$
$$x(3) = 10\cos(3\pi/2) = \mathbf{0}$$

Previous Exams



Rev. Sheet Q9 (a) — periodicity: x(t) is a continuous-time sinusoid, so it is periodic for every f:

$$T_p = 1/f = 1/750 \text{ sec} = \mathbf{1.333 \text{ msec}}$$

Result fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0

Key idea نوع M1: f = 750 Hz و Fs = 3000 أكبر من 2f، فالإشارة ترجع. F = 1/4 يعني Np = 4، والـ samples هي 10 و0 و−10 و0.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 12-1-2019 · Final 28-12-2019 · Revision Sheet 1 · Q9

P10 3×

x(t) = 10 cos(1000πt), Fs = 2000 sample/s

For x(t) = 10 cos(1000πt):

(i) If sampled with Fs = 2000 sample/s, is it possible to reconstruct it successfully?

(ii) Find the Nyquist frequency F_N.

(iii) Using F_N, find an expression for x(n).

(iv) Find the values of x(n) in the first cycle.

$$x(t) = 10\cos(1000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1000\pi \rightarrow \mathbf{f = 500\ Hz}$$

(i), (ii)

$$F_N = 2f_{max} = 2 \times 500 = \mathbf{1000\ Hz}$$
$$Fs = 2000\ Hz \geq 2f_{max} = 1000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(iii) Using the obtained Nyquist frequency F_N = 1000 Hz:

If the signal is sampled at F_N = 1000 Hz (t = nT = n/Fs), the discrete-time signal is:

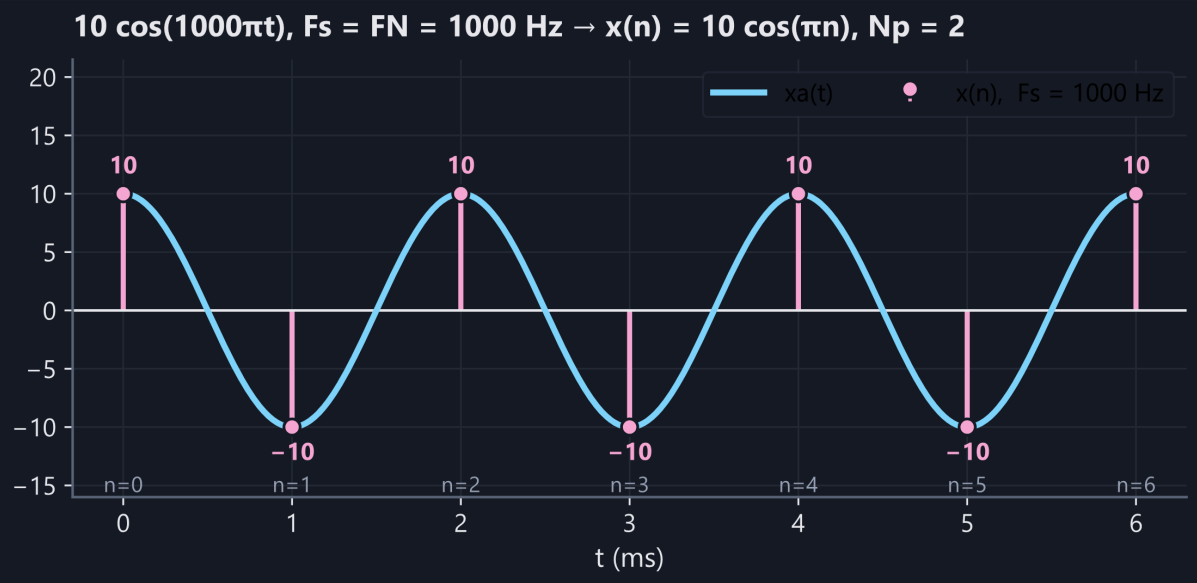
$$x(n) = xa(nT) = xa(n/Fs) = 10\cos(1000\pi\, n/1000) = \mathbf{10\cos(\pi\, n)}$$

$$F = f/Fs = 500/1000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi) = \mathbf{-10}$$

Previous Exams



Result `fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10`

Key idea نوع M1 بس المطلوب نستخدم الـ Nyquist frequency: FN = 2f = 1000 Hz، فـ F = 1/2 و Np = 2، والـ samples بتطلع ±10 على القمم بس.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 11-6-2023 · Q1-b · Final 20-1-2025 · Q3-c · Final 23-1-2023 · Q4-b

P11 ● Frequent 4×

$x(t) = A\cos(2\pi f_o t)$, $A = 2$ V, $f_o = 10$ kHz

For $x_a(t) = A\cos(2\pi f_o t)$, $A = 2$ volt, $f_o = 10$ kHz:

a) Check the periodicity and find the fundamental period.

b) If sampled at Fs = 5 kHz, is it possible to reconstruct the original signal?

c) What is the minimum sampling frequency?

d) Using it, find x(n) and the sample values during one cycle (or the first four samples).

$$x_a(t) = A\cos(2\pi f_0 t),\ A = 2\text{ V},\ f_0 = 10\text{ kHz} \rightarrow x_a(t) = 2\cos(20000\pi t)$$

a) For every fixed value of f, xa(t) is periodic: **$T_p = 1/f_o = 1/10^4 = 0.1$ msec**.

b)

$$F_N = 2f_{max} = 2 \times 10 = \mathbf{20\text{ kHz}}$$
$$Fs = 5\text{ kHz} < 2f_{max} = 20\text{ kHz}$$

Since $Fs < 2f_{max}$, aliasing occurs and the signal **cannot be reconstructed** from its samples.

c) The minimum sampling frequency is **$Fs = 2f_{max} = 20$ kHz**.

d)

If the signal is sampled at Fs = 20000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(20000\pi\ n/20000) = \mathbf{2\cos(\pi\ n)}$$

$$F = f/Fs = 10000/20000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$

First four samples: x(0) = 2, x(1) = −2, x(2) = 2, x(3) = −2

Previous Exams



Result `Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2`

Key idea Fs = 5 kHz أقل من 2f = 20 kHz فيحصل aliasing. أقل Fs يرجّع الإشارة هو 20 kHz، وعنده كل الـ samples بتطلع ±2.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 9-6-2026 · Q1-b · Final 17-8-2026 · Q1-b · Quiz 1 Summer 26 · Q2 · Revision Sheet 1 · Q10

P12 2×

x(t) = 2 cos(2000πt), Fs = 3000 Hz then Fs = 8000 Hz

Given x(t) = 2 cos(2000πt) to be sampled with Fs = 3000 Hz:

(i) Could the original signal be reconstructed successfully from its samples?

(ii) Take Fs = 8000 Hz; find x(n) and its fundamental period N_P.

(iii) Find the sample values of the first cycle and plot x(n).

$$x(t) = 2\cos(2000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 2,\quad 2\pi f = 2000\pi \rightarrow \mathbf{f = 1000\ Hz}$$

(i)

$$F_N = 2f_{max} = 2 \times 1000 = \mathbf{2000\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 2000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(ii), (iii)

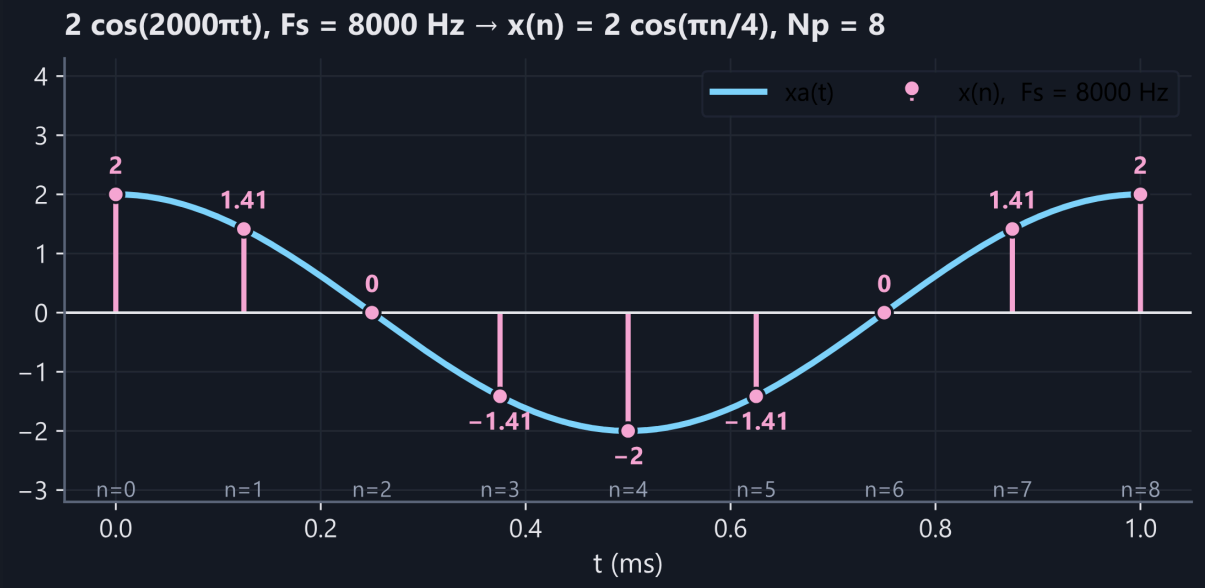
If the signal is sampled at Fs = 8000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(2000\pi\, n/8000) = \mathbf{2\cos(\pi/4\, n)}$$

$$F = f/Fs = 1000/8000 = \mathbf{1/8} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 8\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 7):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi/4) = \mathbf{1.414}$$
$$x(2) = 2\cos(\pi/2) = \mathbf{0}$$
$$x(3) = 2\cos(3\pi/4) = \mathbf{-1.414}$$
$$x(4) = 2\cos(\pi) = \mathbf{-2}$$
$$x(5) = 2\cos(5\pi/4) = \mathbf{-1.414}$$
$$x(6) = 2\cos(3\pi/2) = \mathbf{0}$$
$$x(7) = 2\cos(7\pi/4) = \mathbf{1.414}$$



Result No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414

Key idea مع Fs = 3000 الإشارة ترجع لأنها أكبر من 2f = 2000. ومع Fs = 8000 بيبقى F = 1/8 يعني Np = 8 samples في الـ cycle.

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Midterm 12-11-2025 · Q1 · Midterm Solve 14-11-2019 · Q1-c

P14 3×

Is Fs suitable? (Fs = Nyquist rate exactly)

(a) x(t) = 2 cos(3000πt), Fs = 3 kHz. (b) x(t) = 2 cos(2000πt), Fs = 2 kHz. (c) x(t) = cos(300πt), Fs = 300 Hz.

Is Fs suitable? If so, find x(n), the values in the first cycle, and plot.

(a)

Previous Exams

$$x(t) = 2\cos(3000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 2, \quad 2\pi f = 3000\pi \rightarrow \mathbf{f = 1500\ Hz}$$

$$F_N = 2f_{max} = 2 \times 1500 = \mathbf{3000\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 3000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

$Fs = 2f_{max}$ exactly (the Nyquist rate): it is the minimum rate. Sampling at a rate higher than the Nyquist rate is preferred (Example 1.2).

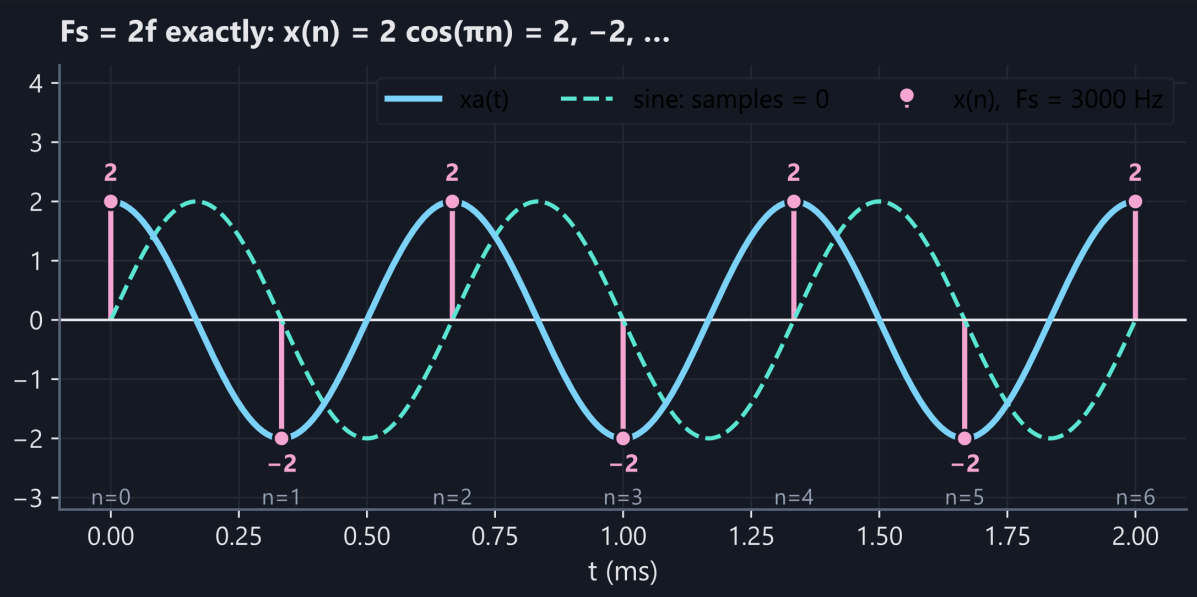
If the signal is sampled at Fs = 3000 Hz ($t = nT = n/Fs$), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(3000\pi\, n/3000) = \mathbf{2\cos(\pi n)}$$

$$F = f/Fs = 1500/3000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$



(b) x(t) = 2 cos(2000πt), Fs = 2 kHz → f = 1000 Hz, Fs = 2f → x(n) = 2 cos(πn): 2, −2.

(c) x(t) = cos(300πt), Fs = 300 Hz → f = 150 Hz, Fs = 2f → x(n) = cos(πn): 1, −1.

Result Fs = 2fmax exactly → x(n) = 2 cos(πn): 2, −2 (works, but a rate above Nyquist is preferred)

Key idea لما Fs = 2f بالظبط الـ samples بتقع على القمم بس (A±). ده ينفع نظريًا، بس الأحسن نعمل sampling أسرع من الـ Nyquist rate.

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Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Final 2-1-2017 · Q1-c · Final 23-12-2017 · Q1-c · Book Midterm 2015 Model-A

P15 1×

Two signals with two sampling frequencies

x1(t) = 5 cos 200πt and x2(t) = 2 sin 500πt are sampled with Fs1 = 400 Hz and Fs2 = 1 kHz respectively.
(i) Are the sampling frequencies suitable? (ii) If suitable, find x1(n) and x2(n).

x1(t):

$$x1(t) = 5\cos(200\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 5,\quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100\ Hz}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200\ Hz}$$
$$Fs = 400\ Hz \geq 2f_{max} = 200\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 5\cos(200\pi\, n/400) = \mathbf{5\cos(\pi/2\, n)}$$

$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 4\ samples}$$

x2(t):

$$x2(t) = 2\sin(500\pi t) \rightarrow 2\pi f = 500\pi \rightarrow \mathbf{f = 250\ Hz}$$

$$F_N = 2f_{max} = 2 \times 250 = \mathbf{500\ Hz}$$
$$Fs = 1000\ Hz \geq 2f_{max} = 500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

If the signal is sampled at Fs = 1000 Hz ($t = nT = n/Fs$), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 2\sin(500\pi\, n/1000) = \mathbf{2\sin(\pi/2\, n)}$$

$$F = f/Fs = 250/1000 = \mathbf{1/4} = k/N_p \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_p = 4\ samples}$$

∴ Both sampling frequencies are suitable.

Result x1: Np = 4 · x2: Np = 4 · both sampling frequencies are suitable (no aliasing)

Key idea كل إشارة نشيكها لوحدها مع الـ Fs بتاعها. الاتنين Fs بتاعهم أكبر من أو يساوي 2f، فالاتنين ينفعوا.

Key words x1(t) x2(t)

Source — from your PDFs

Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Came in	Book Midterm 2015 Model-B

P17 1×

x(n) = 2 cos(nπ/4): periodicity and first four samples

For x(n) = 2 cos(nπ/4): a) Check the periodicity of this signal and find the fundamental period. b) Calculate the sample values for the first four samples.

(a) x(n) = 2 cos(πn/4) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$A = 2,\ \theta = 0 \quad \cdot \quad 2\pi Fn = \pi n/4 \rightarrow F = \mathbf{1/8} = k/N_P \text{ (k and N relatively prime)} \rightarrow \text{rational}$$

$$x(n + 8) = 2\cos(\pi(n + 8)/4) = 2\cos(\pi n/4 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N_P = 8 samples**.

(b) The first four samples (n = 0, 1, 2, 3):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi/4) = \mathbf{1.414}$$
$$x(2) = 2\cos(\pi/2) = \mathbf{0}$$
$$x(3) = 2\cos(3\pi/4) = \mathbf{-1.414}$$

Result F = 1/8 → N = 8 samples; x(0..3) = 2, 1.414, 0, −1.414

Key idea الإشارة discrete أصلًا: F = 1/8 rational فهي periodic و N = 8، والـ samples نطلعها بالتعويض بـ n = 0 و1 و2 و3.

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Came in Revision Sheet 1 · Q11

★ Cheat sheet · Previous Exams

Final results

P7	Tp = 2π/5 s · N = 2 · N = 200 · N = 70 · Tp = 1/1500 s · Tp = 5 ms · sin(2n + π/4) not periodic
P8	fN = 200 Hz, no aliasing · x(n) = cos(πn/2), Np = 4: 1, 0, −1, 0
P9	fN = 1500 Hz, no aliasing · x(n) = 10 cos(πn/2), Np = 4: 10, 0, −10, 0
P10	fN = 1000 Hz, no aliasing · at FN: x(n) = 10 cos(πn), Np = 2: 10, −10
P11	Fs = 5 kHz < 20 kHz → aliasing · Fs,min = 20 kHz · x(n) = 2 cos(πn): 2, −2, 2, −2
P12	No aliasing · Fs = 8 kHz: x(n) = 2 cos(πn/4), Np = 8: 2, 1.414, 0, −1.414, −2, −1.414, 0, 1.414
P14	Fs = 2fmax exactly → x(n) = 2 cos(πn): 2, −2 (works, but a rate above Nyquist is preferred)
P15	x1: Np = 4 · x2: Np = 4 · both sampling frequencies are suitable (no aliasing)
P17	F = 1/8 → N = 8 samples; x(0..3) = 2, 1.414, 0, −1.414

SECTION 07

Sheet 1 Theory

Questions 1.1 to 1.5

Questions

☐	1.1	Define: Signal, System, Signal processing (state one example for each).	p. 60
☐	1.2	Classify the different types of signal processing.	p. 61
☐	1.3	Classify the different signal types.	p. 62
☐	1.4	Explain the basic three operations performed in A/D converters.	p. 63
☐	1.5	State and explain the sampling theorem.	p. 64

Sheet 1 Theory

1.1

Define: Signal, System, Signal processing (state one example for each).

1.1 Define the following items: Signal, system, signal processing. (State one example for each).

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: $s1(t) = 5t$ (varies linearly with t), $s2(t) = 20t^2$ (varies quadratically with t), and $s(x, y) = 3x + 2xy + 10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

Key idea إننا نعدّي الـ Processing جهاز بيشتغل عليها، والـ System كمية بتتغير، والـ Signal على الـ system.

Key words: Signal | System | Signal processing

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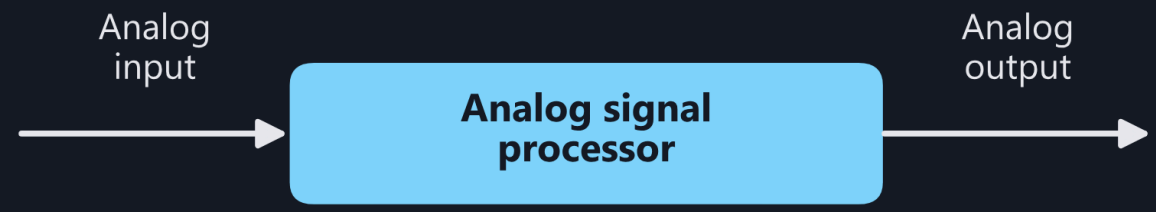
Lecture	**LEC 2** · Sec. 1.1 · p. 4
Sheet	**Sheet 1** · Q 1.1

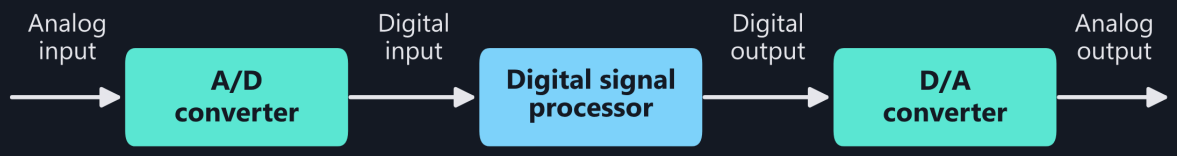
1.2

Classify the different types of signal processing.

There are two types of signal processing systems (Sec. 1.2):

1. **Analog signal processor (Fig. 1.1):** the analog input signal is processed directly by an analog signal processor to give an analog output signal.
2. **Digital signal processor (Fig. 1.2):** provides an alternative method for processing the analog signal. The **A/D converter** converts the analog input into a digital signal (a sequence of numbers), the **digital signal processor** performs the required operations on it, and the **D/A converter** converts the result back into an analog output signal.





Key idea نوعين: analog signal processing بيشتغل على الإشارة على طول، و digital signal processing بيعدّيها على A/D ثم DSP ثم D/A.

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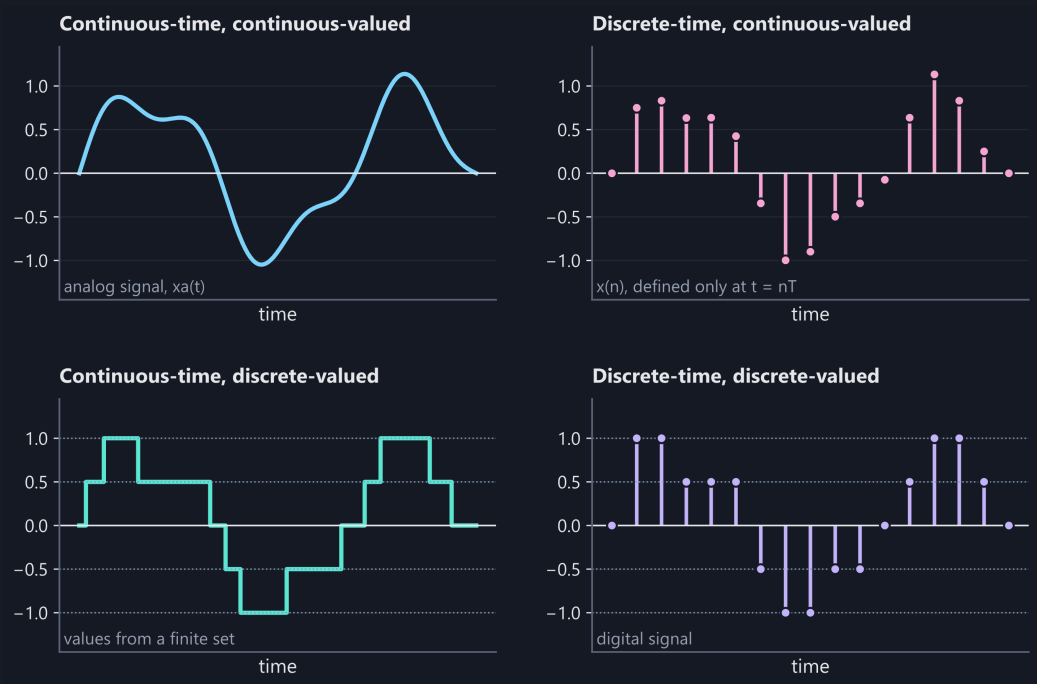
Lecture	**LEC 2** · Sec. 1.2 · p. 11
Sheet	**Sheet 1** · Q 1.2

Sheet 1 Theory

1.3

Classify the different signal types.

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.
4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.



Key idea نفس تصنيفات الإشارة كأزواج: continuous / discrete time، و continuous / discrete valued، و deterministic / random، و multichannel / multidimensional.

Key words Deterministic vs. random

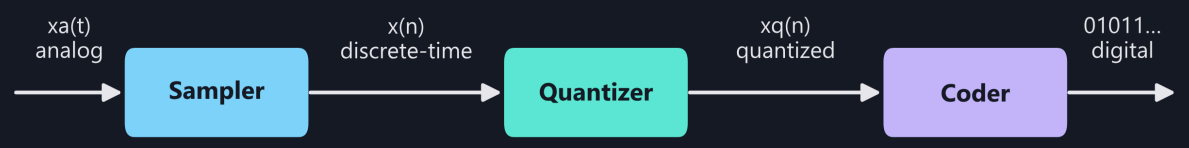
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Lecture	**LEC 2** · Sec. 1.4 · p. 18
Sheet	**Sheet 1** · Q 1.3

1.4

Explain the basic three operations performed in A/D converters.

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

Key idea sampling بعده quantization بعده coding: نقطّع الوقت، وبعدين القيم، وبعدين نحوّلها binary.

Sheet 1 Theory

Key words Sampling Quantization Coding

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Lecture	**LEC 4** · Sec. 1.6 · p. 3
Sheet	**Sheet 1** · Q 1.4

1.5

State and explain the sampling theorem.

- To select Fs we must have information about the frequency content of the signal, e.g. f_{max} = 3000 Hz for speech and 5 MHz for television signals. The signal may be passed through a filter that severely attenuates components above f_{max}.
- To avoid aliasing, Fs is selected so that **$Fs > 2f_{max}$** (1.47). Then every component $|f_i| < f_{max}$ is mapped into $-1/2 \le F_i = f_i/Fs \le 1/2$ (1.48).

Sampling theorem: if the highest frequency contained in an analog signal xa(t) is $f_{max} = B$ and the signal is sampled at a rate $Fs > 2f_{max} = 2B$, then xa(t) can be exactly recovered from its sample values using the interpolation function:

$$g(t) = \frac{\sin(2\pi Bt)}{(2\pi Bt)} \quad (1.50) \qquad xa(t) = \sum xa(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

- The sampling rate **$f_N = 2B = 2f_{max}$** is called the **Nyquist rate**.
- Because of the complexity and the infinite number of samples required, these reconstruction formulas are primarily of theoretical interest.

Key idea لو Fs > 2fmax نقدر نرجّع الإشارة الأنالوج بالظبط من الـ samples بتاعتها، والـ Nyquist rate هو 2fmax.

Key words Sampling theorem

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Lecture	**LEC 4** · Sec. 1.6.2 · p. 17
Sheet	**Sheet 1** · Q 1.5

★ Cheat sheet · Sheet 1 Theory

Remember

1.1 Signal = varying physical quantity; system = device that operates on it; processing = passing the signal through the system.

1.2 Analog signal processing (direct) and digital signal processing (A/D → DSP → D/A).

1.3 Continuous/discrete time, continuous/discrete valued, deterministic/random, multichannel/multidimensional.

1.4 Sampling → quantization → coding.

1.5 If Fs > 2 fmax the analog signal can be recovered exactly from its samples (Nyquist rate = 2 fmax).

Equations

$$g(t) = \sin(2\pi Bt) / (2\pi Bt) \quad (1.50)$$

$$xa(t) = \Sigma\, xa(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

SECTION 08

Sheet 1 Problems

Questions 1.6 to 1.8

Questions

- ☐ **1.6** Which sinusoids are periodic? Compute their fundamental period. p. 66
- ☐ **1.7** xa(t) = 3 sin(100πt): sketch, sample at Fs = 300, period, peak value p. 67

☐ 1.8 A/D with T1 = 5 ms then D/A with T2 = 1 ms: find ya(t) p. 70

1.6 ● Frequent 4×

Which sinusoids are periodic? Compute their fundamental period.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.
(a) xa(t) = 3 cos(5t + π/6) (b) xa(t) = 3 cos(100πt) (c) x(n) = cos[π(30n/105)] (d) x(n) = 2 exp[j(n/6 − π)]

(a)

xa(t) = 3 cos(5t + π/6) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 2\pi/5 \text{ sec}}$$
$$x_a(t + T_p) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

∴ Periodic, fundamental period Tp = 2π/5 sec.

(b)

xa(t) = 3 cos(100πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 100\pi \rightarrow f = 50 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 20 \text{ msec}}$$
$$x_a(t + T_p) = 3\cos(100\pi(t + 20 \text{ ms})) = 3\cos(100\pi t + 2\pi) = x_a(t)$$

∴ Periodic, fundamental period Tp = 20 msec.

(c)

x(n) = cos[π(30n/105)] is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 30\pi/105 \rightarrow F = 1/7 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 7) = \cos(30\pi(n + 7)/105) = \cos(30\pi n/105 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 7 samples**.

(d) x(n) = 2 exp[j(n/6 − π)] = $2e^{-j\pi}\, e^{j2\pi Fn}$: a discrete-time complex exponential is periodic only if its relative frequency is rational.

$$2\pi F = 1/6 \rightarrow F = 1/(12\pi)$$

F cannot be expressed as a ratio of two integers → **not periodic**.

Result Tp = 2π/5 s · Tp = 20 ms · N = 7 samples · 2 exp[j(n/6 − π)] not periodic

Key idea continuous الـ دايمًا periodic. discrete الـ نشيك F: لو rational تبقى periodic والمقام بعد الاختصار هو N، ولو فيها π تبقى مش periodic.

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Lecture	**LEC 3** · Sec. 1.5.1 · p. 3
Lecture	**LEC 3** · Sec. 1.5.2 · p. 9
Sheet	**Sheet 1** · Q 1.6

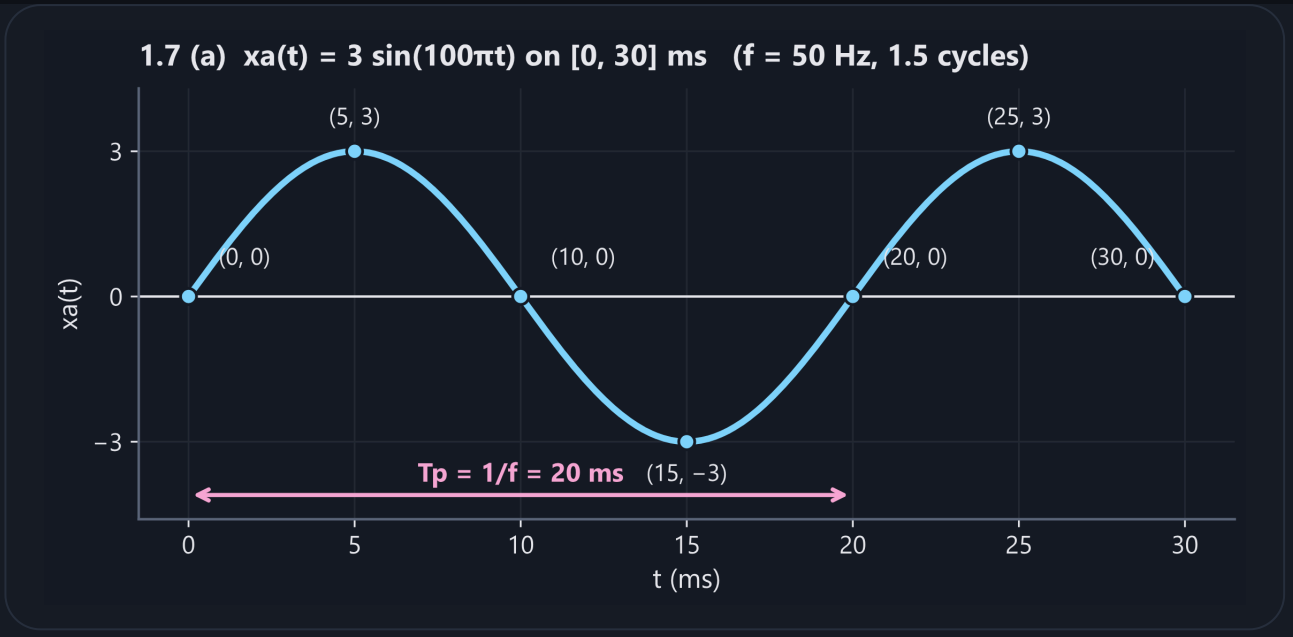
1.7 ● Frequent 8×

xa(t) = 3 sin(100πt): sketch, sample at Fs = 300, period, peak value

1.7 Consider the analog sinusoid signal xa(t) = 3 sin(100πt).
(a) Sketch the signal for [0, 30] ms.
(b) The signal is sampled with sampling rate Fs = 300 samples per second. Determine the frequency of the discrete-time signal and show that it is periodic.
(c) Compute the sample values in one period of x(n). Sketch x(n) on the same diagram with xa(t). What is the period of this discrete-time signal x(n)?
(d) Can you find a sampling rate Fs such that the signal x(n) reaches its peak value 3? What is the minimum Fs suitable for this task?

(a) xa(t) = 3 sin(100πt) = A sin(2πft): A = 3, 2πf = 100π → f = 50 Hz, Tp = 1/f = 20 msec. The interval [0, 30] ms contains 1.5 periods.

Sheet 1 Problems



(b)

If the signal is sampled at Fs = 300 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 3\sin(100\pi\, n/300) = \mathbf{3\sin(\pi/3\, n)}$$

$$F = f/Fs = 50/300 = \mathbf{1/6} = k/N_p \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_p = 6 \text{ samples}}$$

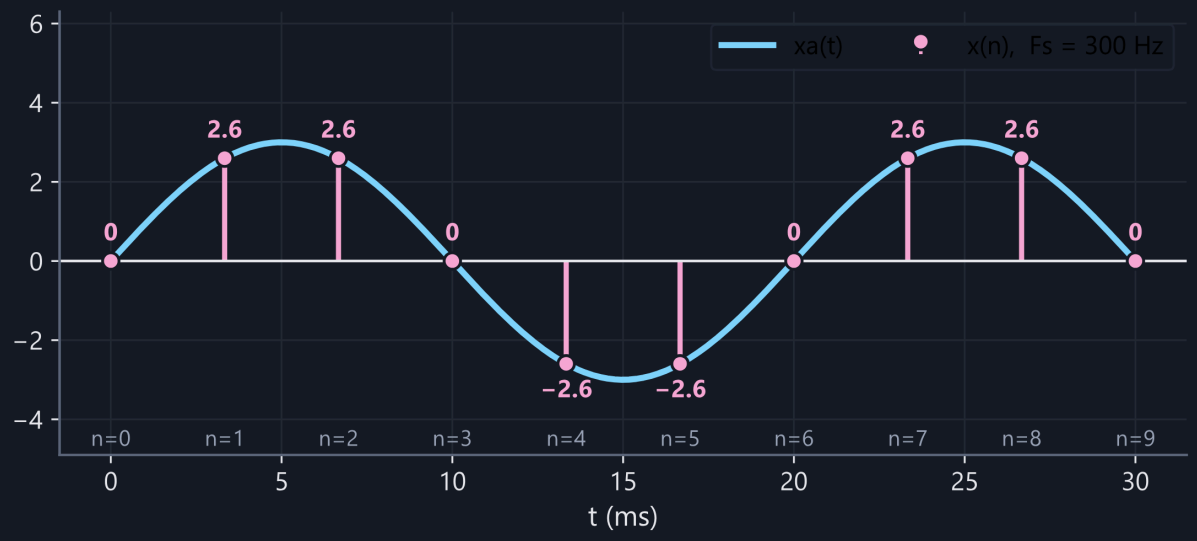
$$x(n+6) = 3\sin(\pi(n+6)/3) = 3\sin(\pi n/3 + 2\pi) = x(n)$$

(c)

$$x(0) = 3\sin(0) = \mathbf{0}$$
$$x(1) = 3\sin(\pi/3) = \mathbf{2.598}$$
$$x(2) = 3\sin(2\pi/3) = \mathbf{2.598}$$
$$x(3) = 3\sin(\pi) = \mathbf{0}$$
$$x(4) = 3\sin(4\pi/3) = \mathbf{-2.598}$$
$$x(5) = 3\sin(5\pi/3) = \mathbf{-2.598}$$

The period of x(n) is **N = 6 samples** = 6 × (1/300) s = 20 msec.

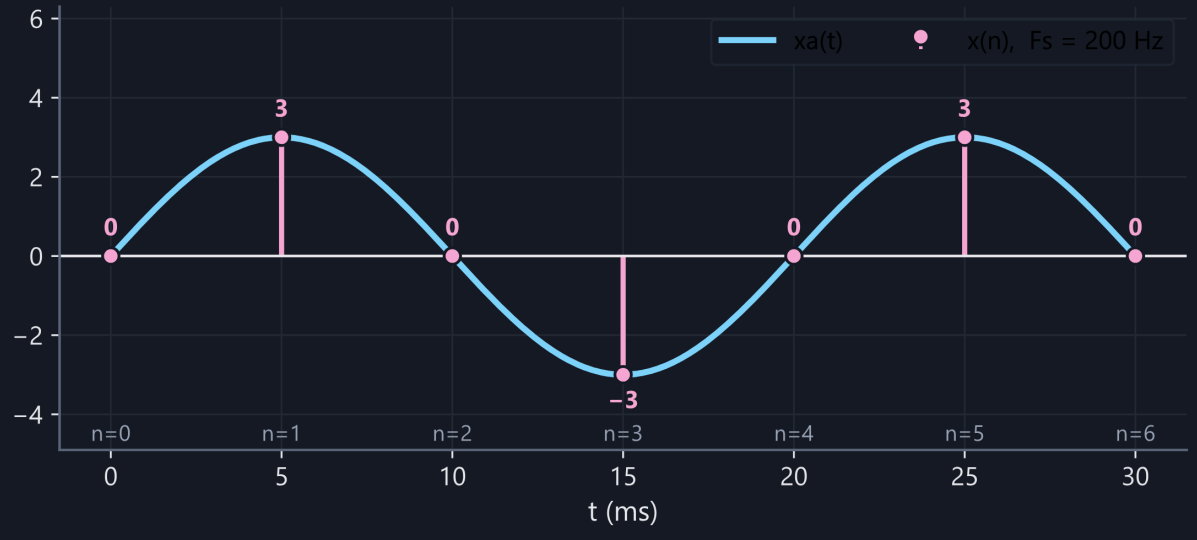
Sheet 1 Problems



(d) x(n) = 3 sin(100πn/Fs) reaches its peak value 3 when:

$$\sin(100\pi n/Fs) = 1 \rightarrow 100\pi n/Fs = \pi/2 \rightarrow Fs = 200n$$

Yes. The minimum is at n = 1: **Fs = 200 samples/sec**. Then x(n) = 3 sin(πn/2) = 0, 3, 0, −3.



Result x(n) = 3 sin(πn/3): 0, 2.598, 2.598, 0, −2.598, −2.598 · N = 6 (20 ms) · Fs = 200 → 0, 3, 0, −3

Key idea f = 50 Hz و Fs = 300، فـ F = 1/6 و N = 6. عشان x(n) يوصل للقمة 3 لازم sample يقع عند ربع الـ period، وده بيحصل لما Fs = 200n.

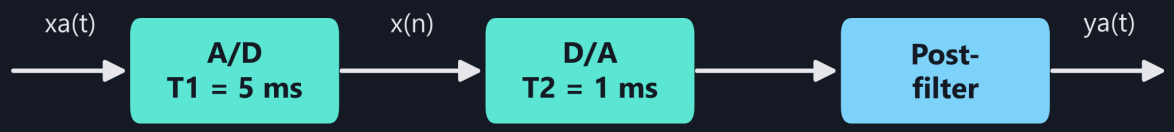
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Lecture	**LEC 4** · Sec. 1.6.1 · p. 8
Sheet	**Sheet 1** · Q 1.7
Came in	Midterm 28-11-2021 · Q1-b · Midterm 2 · Q1-c · Midterm 8-12-2020 · Q1-c · Final 4-3-2021 · Q1-b · Sheet 1 · 1.7

1.8 · Frequent 5×

A/D with T1 = 5 ms then D/A with T2 = 1 ms: find ya(t)

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T1 = 5 ms and T2 = 1 ms respectively. Determine the output ya(t) if the input is given by: xa(t) = 3 cos(100πt) + 2 sin(250πt) (t in second).



Solution

The frequencies existing in the analog signal are f1 = 50 Hz (2πf1 = 100π) and f2 = 125 Hz (2πf2 = 250π).

Sheet 1 Problems

The A/D converter has T1 = 5 ms, so Fs = 1/T1 = 200 samples/s and the folding frequency is Fs/2 = 100 Hz, the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.30) we obtain:

$$
\begin{aligned}
x(n) &= x_a(nT1) = x_a(n/Fs) \\
&= 3\cos 2\pi(50/200)n + 2\sin 2\pi(125/200)n \\
&= 3\cos 2\pi(1/4)n + 2\sin 2\pi(5/8)n \\
&= 3\cos 2\pi(1/4)n + 2\sin 2\pi(1-3/8)n \\
&= 3\cos 2\pi(1/4)n + 2\sin 2\pi(-3/8)n
\end{aligned}
$$

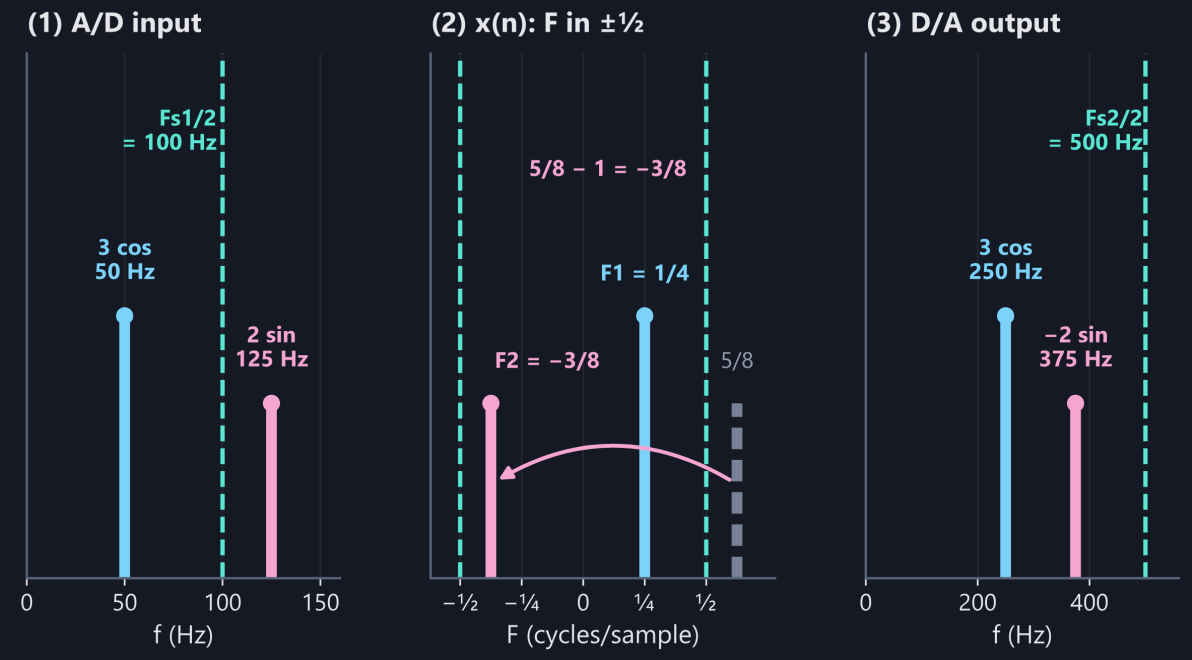
Finally, we obtain:

$$\mathbf{x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)}$$

The D/A converter has T2 = 1 ms (Fs = 1000 samples/s). Using t = nT2, i.e. n = t/T2 = 1000t:

$$y_a(t) = 3\cos(\pi(1000t)/2) - 2\sin(3\pi(1000t)/4) = \mathbf{3\cos(500\pi t) - 2\sin(750\pi t)}$$

which is different from $x_a(t)$: the distortion is caused by aliasing (Fs = 200 Hz < 2 × 125 Hz), and the frequencies are scaled by T1/T2 = 5.



Sheet 1 Problems

Result `x(n) = 3 cos(πn/2) − 2 sin(3πn/4) · ya(t) = 3 cos(500πt) − 2 sin(750πt) (aliasing)`

Key idea الـ A/D بيعمل sampling بـ Fs = 200 Hz، فالـ 125 Hz بتعمل alias. والـ D/A بيطلّع الـ samples أسرع 5 مرات (T2 = 1 ms)، فكل الترددات بتتضرب في 5.

Source — from your PDFs

Lecture	**LEC 4** · Example 1.3 · p. 44
Sheet	**Sheet 1** · Q 1.8
Came in	Revision Sheet 1 · Q12 · Sheet 1 · 1.8

★ Cheat sheet · Sheet 1 Problems

Final results

- **1.6** Tp = 2π/5 s · Tp = 20 ms · N = 7 samples · 2 exp[j(n/6 − π)] not periodic
- **1.7** x(n) = 3 sin(πn/3): 0, 2.598, 2.598, 0, −2.598, −2.598 · N = 6 (20 ms) · Fs = 200 → 0, 3, 0, −3
- **1.8** x(n) = 3 cos(πn/2) − 2 sin(3πn/4) · ya(t) = 3 cos(500πt) − 2 sin(750πt) (aliasing)